The American Association of Endocrine Surgeons Guidelines for the Definitive Surgical Management of Secondary and Tertiary Renal Hyperparathyroidism

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Objective: To develop evidence-based recommendations for safe, effective, and appropriate treatment of secondary (SHPT) and tertiary (THPT) renal hyperparathyroidism.

Background: Hyperparathyroidism is common among patients with chronic kidney disease, end-stage kidney disease, and kidney transplant. The surgical management of SHPT and THPT is nuanced and requires a multidisciplinary approach. There are currently no clinical practice guidelines that address the surgical treatment of SHPT and THPT.

Methods: Medical literature was reviewed from January 1, 1985 to present January 1, 2021 by a panel of 10 experts in SHPT and THPT. Recommendations using the best available evidence was constructed. The American College of Physicians grading system was used to determine levels of evidence. Recommendations were discussed to consensus. The American Association of Endocrine Surgeons membership reviewed and commented on preliminary drafts of the content.

Results: These clinical guidelines present the epidemiology and pathophysiology of SHPT and THPT and provide recommendations for work-up and management of SHPT and THPT for all involved clinicians. It outlines the preoperative, intraoperative, and postoperative management of SHPT and THPT, as well as related definitions, operative techniques, morbidity, and outcomes. Specific topics include Pathogenesis and Epidemiology, Initial Evaluation, Imaging, Preoperative and Perioperative Care, Surgical Planning and Parathyroidectomy, Adjuncts and Approaches, Outcomes, and Reoperation.

Conclusions: Evidence-based guidelines were created to assist clinicians in the optimal management of secondary and tertiary renal hyperparathyroidism.

Keywords: hyperparathyroidism, kidney, parathyroid, parathyroidectomy, pathogenesis, postoperative care, preoperative, renal, secondary, tertiary

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Hyperparathyroidism (HPT) in the setting of kidney disease develops as a physiologic adaptation to alterations in calcium, vitamin D, and fibroblast growth factor 23 (FGF23) levels. Elevations in parathyroid hormone (PTH) level are initially compensatory but become pathologic, resulting in hypercalcemia, renal osteodystrophy, and cardiovascular morbidity, in addition to other manifestations. The treatment of secondary hyperparathyroidism (SHPT) and tertiary hyperparathyroidism (THPT) begins with medical management but often necessitates parathyroidectomy (PTX) for definitive treatment to remove the source of the excess PTH. The management of both SHPT and THPT calls for a nuanced, multidisciplinary approach given the complex physiologic alterations results from the underlying kidney disease.

Recognizing the need for guidance in the management of this nuanced pathophysiologic process, the American Association of Endocrine Surgeons (AAES) determined that evidence-based clinical guidelines are necessary to enhance the safe and effective surgical treatment of secondary and tertiary renal HPT A multidisciplinary panel was convened with representation from the disciplines of nephrology and surgery to supply recommendations for all involved clinicians. These guidelines aim to:

(1) Assist providers in their understanding of the epidemiology and pathogenesis of kidney-related parathyroid disease.
(2) Describe the diagnosis of kidney-related HPT.
(3) Provide an overview of the nonoperative management of kidney-related HPT.
(4) Define the indications for surgical intervention and the extent of surgery.
(5) Detail methods for safe and effective perioperative management.

The diagnosis, treatment, and management of primary hyperparathyroidism (PHPT) and medical management of secondary renal HPT has been addressed by national organizations and will not be addressed in this document.[1–4] We recognize the

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tremendous efforts that have been put into these previous works and did not attempt to replicate them here, we instead recommend they be reviewed by the readers of this document as needed. The diagnosis and management of SHPT and THPT resulting from etiologies not related to kidney disease are outside the scope of this document. The guidelines presented here focus on surgical management and with the goal of achieving definitive surgical treatment of kidney-related parathyroid disease in a safe and effective manner.

METHODS

Writing Group

The AAES approved this project at its' initiation. A multidisciplinary panel was appointed by the AAES President to include individuals with expertise in kidney-related HPT. Individual writing subcommittees were created for each of the 7 topic sections outlined, groups were comprised of 2 to 3 co-authors. Topic outlines and drafts were discussed to consensus.

Literature Evidence, Recommendations, and Clinical Highlights

Medical literature search parameters were set from January 1, 1985 to January 1, 2021. Landmark articles or articles that provide historical perspective which may predate the search parameters were allowed at the authors' discretion. The primary coauthors for each subsection conducted a PubMed subject search using Boolean logic for relevant terms. Selected publications were limited to full manuscripts in the English language appearing in core journals. In addition to clinical recommendations, the writing group chose to include clinical highlights to emphasize summative points of interest.

Grading and Practice Recommendations

This manuscript was prepared using the 2010 American College of Physicians Grading system to evaluate the strength and quality of evidence[5] (https://www.acpjournals.org/doi/10.7326/0003-4819-153-3-201008030-00010).

Formal feedback and suggestions were sought from non-author AAES and American Society of Nephrology experts. The input was also sought from the AAES membership by online survey in January 2021; the input was incorporated from the above sources. These guidelines also underwent a standard peer-review process whereby 4 anonymous reviewers were selected by the journal editor, revisions were made, and responses provided.

American Association of Endocrine Surgeons (AAES) and American Society of Nephrology (ASN) Endorsement

Feedback and suggestions were discussed, and consensus revisions made as needed. The document was reviewed and endorsed by both organizations.

Cautions to Implementation

These guidelines present the authors' consensus view of the management of renal HPT based on the current available evidence. The responsibility of maintaining an updated working knowledge for the care of renal HPT is the responsibility of treating physicians as the current evidence will change. The guidelines are not intended for use as a basis to approve or deny financial coverage for any therapeutic or diagnostic modality; cannot account for individual patient characteristics; may not address all relevant comorbidities, subpopulations, effects on patient quality of life, or other factors; cannot be considered inclusive of all proper methods of care; and may omit other treatments reasonably directed at obtaining the same results. The authors accept no responsibility for inaccuracies or information perceived as misleading and recommend users independently confirm the information presented is correct. The guidelines are meant to be flexible and do not account for individual patient characteristics. They do not represent the only approach to treating renal HPT and should not replace individual physician judgment. The content was subject to external peer review.

INITIAL EVALUATION

Pathogenesis of Kidney-related HPT

SHPT is an almost universal phenomenon of chronic kidney disease (CKD) that occurs in ~40% of individuals with CKD grade 3, 82% of individuals with CKD grade 4, and essentially all patients with CKD grade 5 or end-stage kidney disease (ESKD).[6–8] While SHPT is initially a physiologic adaption to progressive CKD, chronic elevation of PTH eventually becomes pathologic. In individuals with normal kidney function, normal serum concentrations of calcium and phosphorus are maintained through the interaction of 3 hormones: PTH, 1, 25-dihydroxycholecalciferol (calcitriol, the active metabolite of vitamin D), and FGF23. The primary physiologic function of PTH is to maintain calcium homeostasis by (1) increasing bone mineral dissolution, thus releasing calcium and phosphorus into the circulation; (2) increasing kidney reabsorption of filtered calcium and excretion of phosphorus; (3) increasing the activity of the kidney CYP27B1 enzyme to convert 25-hydroxyvitamin D to 1,25-dihydroxyvitamin D, or calcitriol, which is the biologically active form of vitamin D; and (4) indirectly enhancing the gastrointestinal absorption of both calcium and phosphorus through its effects on the synthesis of calcitriol. Thus, in individuals with normal kidney function a small decrease in the ionized calcium results in the release of PTH to restore serum calcium and maintain serum phosphorus concentrations. The kidneys are critical to this normal homeostatic response, but in CKD, the kidney loses the capacity to maintain normal phosphorus and calcium homeostasis. As CKD progresses there is decreased production of calcitriol and an increase in FGF23, which ultimately results in the retention of phosphate, a mild decrease in ionized calcium, and an increase in PTH (Fig. 1).[7,9–13]

The extracellular concentration of ionized calcium is the most important determinant of minute-to-minute secretion of PTH from stored secretory granules. The calcium-sensing receptor (CaSR) mediates the rapid response in PTH secretion, which occurs within seconds of changes in ionized calcium concentrations.[14,15] Phosphorus increases PTH production by enhancing the stability of PTH mRNA.[16] Data from mice confirm that FGF23 directly alters PTH release in an alpha-Klotho-independent manner although Klotho may be synergistic with the CaSR in regulating parathyroid gland growth.[17,18] Calcitriol suppresses PTH release via the vitamin D receptor (VDR) to lead to direct gene suppression. Other vitamin D compounds that bind to the VDR with lower affinity still reduce PTH release if given in high enough quantities, including 25(OH)D-calcidiol.[19] The chronic increase in PTH production is associated with an increase in parathyroid gland size due to cell proliferation which leads to an increase in cell number.[20–23] As CKD progresses to ESKD, most patients develop 4-gland enlargement. As noted in Figure 2, the parathyroid glands

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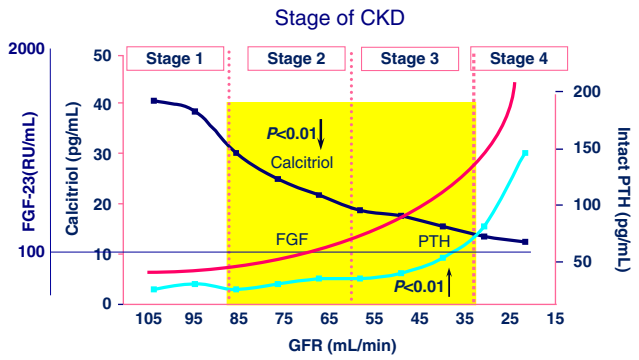

FIGURE 1. Changes in calcitriol, FGF23, and PTH as CKD progresses. Adapted from Gal-Moscovici and Sprague.[9] Adaptations are themselves works protected by copyright. So in order to publish this adaptation, authorization must be obtained both from the owner of the copyright in the original work and from the owner of copyright in the translation or adaptation. GFR indicates glomerular filtration rate.

significantly enlarge from a normal size to be 500 to 1000 mg per gland with nodule formation and remains histologically hypercellular.[25,26] As this process progresses, some glands may descend into deep retroesophageal or paratracheal spaces and enlargement of ectopic parathyroid rests in the cervical thymus can occur.[27,28]

Normal parathyroid glands are comprised predominantly of 2 major cell types, the small chief cells characterized by rather scant, lightly stained cytoplasm, and much larger, eosinophilic oxyphil cells, which are often found in small clusters (Fig. 2). The chief cells are responsible for the production of PTH, while the function of the oxyphil cells is unknown. The increase of the parathyroid parenchymal mass is due to the proliferation of chief cells and oxyphil cells in the presence of a known stimulus for PTH secretion.[29] Gland hyperplasia can be categorized into 2 distinct forms, diffuse hyperplasia and nodular hyperplasia which is a consequence of monoclonal chief cell growth. In initial stages of CKD, parathyroid cell proliferation appears to be diffuse and homogeneous, whereas nodules form within enlarged parathyroid glands in advanced stages of CKD and especially as patients develop ESKD. Compared with normal glands, parathyroid glands in ESKD patients have a preponderance of oxyphilic parathyroid cells, frequently have more nodular hyperplasia, fibrosis with a sclerotic architecture, and dystrophic calcifications.[26,29,30]

With the development of parathyroid gland hyperplasia, the parathyroid nodules reduce expression of both the VDR and CaSR (Fig. 3), thus rendering the tissue less responsive to the circulating concentrations of ionized calcium and calcitriol, leading to progression to refractory, or THPT.[31–33] In contrast to SHPT, THPT is characterized by inappropriately increased levels of PTH despite hypercalcemia that is caused by autonomous function of multiple hypertrophied parathyroid gland nodules refractory to medical management. Most often THPT is seen in patients on long-term dialysis or after kidney transplantation. The CaSR and VDR are both down regulated in hyperplastic parathyroid glands, which likely contributes to resistance to calcitriol, calcimimetics, and calcium, a stage at which PTX is likely necessary.[34–36]

Histopathologic studies performed by Tominaga and colleagues on patients who underwent PTX for THPT due to CKD show asymmetric enlargement, nodularity, and an increased number of oxyphil cells. They found that nodular

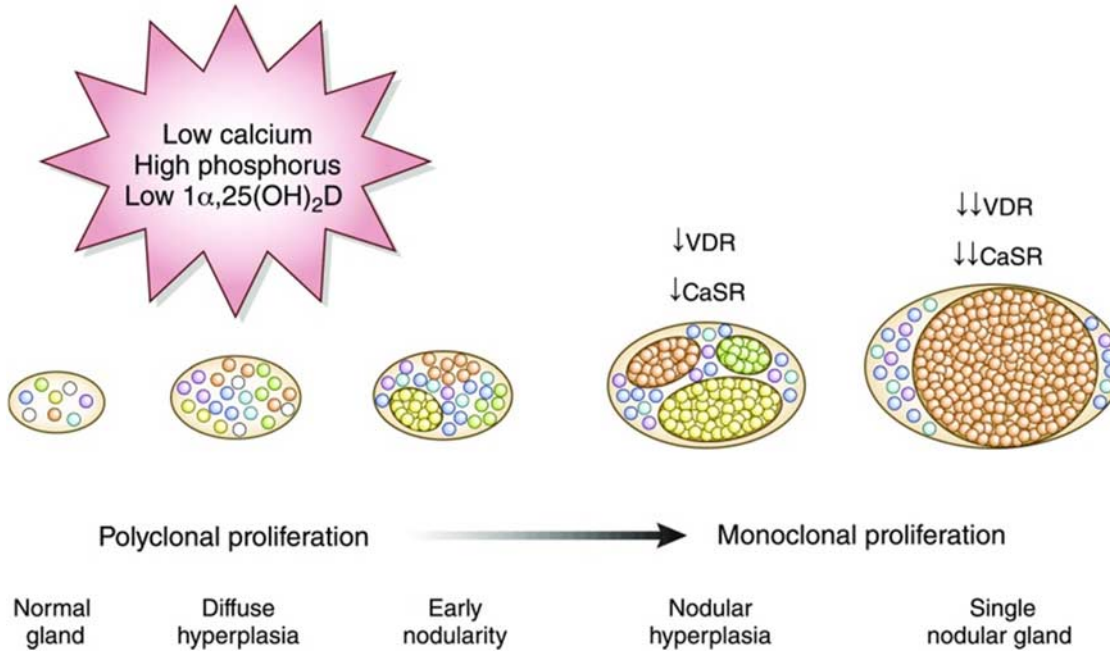

FIGURE 2. Progression of parathyroid hypertrophy with progressive CKD. Adapted from Lau et al.[24] Adaptations are themselves works protected by copyright. So in order to publish this adaptation, authorization must be obtained both from the owner of the copyright in the original work and from the owner of copyright in the translation or adaptation.

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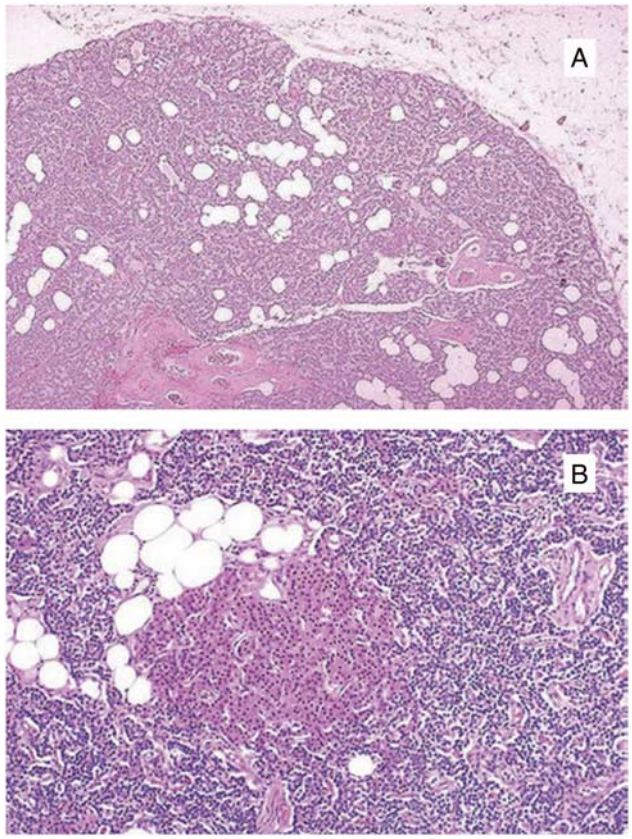

FIGURE 3. Histologic changes seen in secondary hyperparathyroidism. A, Normal parathyroid tissue. B, Parathyroid hyperplasia due to CKD as seen in secondary hyperparathyroidism. Adapted from Cozzolino et al.[15] Adaptations are themselves works protected by copyright. So in order to publish this adaptation, authorization must be obtained both from the owner of the copyright in the original work and from the owner of copyright in the translation or adaptation.

hyperplasia was the most frequent cause of THPT in ESKD patients; they demonstrated the relative number of scattered cells in the synthetic phase was significantly greater in nodular than in diffuse hyperplasia indicating more aggressive proliferation.[25,29] In addition, the calcium set point for the inhibition of PTH secretion was found to be higher in the cells from nodular hyperplasia than in the cells obtained from diffuse hyperplasia.[29] Though there was no difference in expression of PTH mRNA in nodular and diffuse hyperplasia, these data suggest that nodular hyperplasia is more progressively hyperplastic has more aggressive proliferative activities and show more abnormal regulation of PTH secretion.[25,28–30] With parathyroid hyperplasia secondary to CKD, as shown in Figure 3B, all parathyroid glands are enlarged.[30]

Clinical Diagnosis of SHPT

SHPT resulting from underlaying kidney pathology is characterized by diminished serum calcium, elevated serum phosphate, and low 1,25-dihydroxyvitamin D levels in the setting of an elevated PTH. Over the last 20 years, there has been a slow increase in the acceptable range of PTH levels in patients with advanced CKD which has evolved from the growing knowledge of the alterations in PTH function in the setting of CKD. In the 1970s, the blunted effect of PTH in dogs with uremia was identified when Massry et al[37] demonstrated a diminished increase in serum calcium in response to PTH infusion that could be overcome with calcitriol supplementation. Subsequent studies in humans demonstrated a similarly blunted calcemic response to PTH infusion in patients with advanced CKD.[38] These studies led to the hypothesis that elevated PTH values in CKD maintain bone health and raised the question regarding the optimal levels of PTH.

Defining the normal range of PTH in patients with CKD was further complicated by the evolution of PTH immunoassays, as initial immunoassays measured both active as well as inactive fragments due to a lack of kidney excretion. The National Kidney Foundation established the Kidney Disease: Improving Global Outcomes (KDIGO) organization which serves to establish and implement clinical guidelines in kidney disease, including the CKD-Bone and Mineral Disorder guidelines which were published and revised in 2009 and 2017. The first KDIGO guidelines in 2003 defined "normal" PTH range for patients with CKD to be 150 to 300 pg/mL, as the levels that correlated with optimized bone health on bone biopsy. The 2003 KDIGO guidelines further recommended PTX for PTH levels over 1000 pg/mL.[39] Unfortunately, multiple other assays were developed that had different sensitivity and specificity for the active hormone and the various fragments, resulting in variance of ~1.5- to 4.2-fold difference between assays.[40] Thus, subsequent studies on bone composition and PTH levels failed to confirm that a PTH between 150 and 300 pg/mL was optimal.[41]

An increased understanding of the interrelationship of renal osteodystrophy with vascular calcification and cardiovascular pathology shifted the attention of the morbidity of kidney-related HPT from a bone-centric focus to a systemic focus and led to the coining of CKD-Mineral Bone Disorder (CKD-MBD)[42,43] (Table 1). In 2009, the KDIGO guidelines set PTH levels between 2 and 9 times the upper limit of normal (roughly 300–600 pg/mL by most assays), with interventions recommended for levels increasing within the range or levels outside of that range.[44] The upper limit of 600 pg/mL was based on cross sectional studies demonstrating that a significant association between higher PTH levels with increased mortality was detectable only above that threshold.[45,46] These levels were maintained in the 2017 update of the KDIGO guidelines.[2,3]

A large cohort study published in 2020, the Dialysis Outcomes and Practice Patterns (DOPPS) study, examined patients on dialysis for at least 12 months in Europe, Canada, and the United States. Table 2 shows the findings of SHPT and treatments given to suppress PTH. As shown in this table, 17% to 29% patients had PTH levels above the level of 600 pg/mL, the upper level per the KDIGO guidelines.[47] When adjusted for patient age, sex, body mass index, time on dialysis, catheter use, comorbidities, albumin, hemoglobin, creatinine, facility percentage of patients with catheter use, percentage of patients having albumin <3.5 g/dL, and mean hemoglobin, the world wide DOPPS cohort (N = 35,655) demonstrated a hazards ratio (HR) of 1.22 (1.13–1.32) for mortality in patients with intraoperative parathyroid hormone (ioPTH) > 600 pg/mL.

Medical Management of SHPT

Multiple therapeutic options exist for the management of SHPT. The first reports of suppression of PTH and improvement of bone histology with calcitriol was published in the 1980s, and this rapidly became the mainstay of therapy.[48,49] However, due

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TABLE 1. Definition of CKD-Mineral Bone Disorder

A systemic disorder of mineral and bone metabolism due to CKD manifested by either one or a combination of the following:
- Abnormalities of calcium, phosphorus, PTH, or vitamin D metabolism
- Abnormalities in bone turnover, mineralization, volume, linear growth, or strength
- Vascular or other soft tissue calcification

Renal osteodystrophy is an alteration of bone morphology in patients with CKD. It is one measure of the skeletal component of the systemic disorder of CKD-MBD that is quantifiable by histomorphometry of bone biopsy

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to the risk of hypercalcemia, analogues of calcitriol, such as paricalcitol, doxercalciferol, and alphacalcidol, were developed to maximize parathyroid suppression while minimizing intestinal calcium and phosphorus absorption.[50–57] In 2004, cinacalcet, a calcimimetic, that "mimics" the effect of hypercalcemia on CaSR to decrease PTH levels, was FDA approved for use in patient with SHPT and CKD on dialysis, and more recently an intravenous calcimimetic, etalcalcitide was also approved for use.[58–60]

Calcimimetics differ from calcitriol and other vitamin D analogues in that calcium levels are lowered, rather than increased.[61] While there are no large, randomized control trials directly comparing outcomes with the use of the calcimimetics or vitamin D analogues, the EVOLE trial examined cardiovascular risk with cinacalcet compared with placebo. The EVOLE trial randomized hemodialysis patients with a PTH $\geq$300 pg/mL to receive either cinacalcet or a placebo in addition to standard of care, including vitamin D analogues and phosphate binders. The primary endpoint was an unadjusted composite cardiovascular endpoint, without demonstration of superiority and that study did not demonstrate efficacy of cinacalcet [HR 0.93, 95% confidence interval (CI): 0.85–1.02, $P = 0.11$].[62–64] There were many flaws of the study including unequal randomization by age, high drop-in and drop-out rates, and a need to extend the study to achieve the event driven milestones.[65] Subsequent planned secondary analyses of data adjusted for comorbidities did reveal efficacy in prevention of nonatherosclerotic cardiac events and fractures with cinacalcet use and potential reduced need for PTX.[66–68] In 2017, the KDIGO update of CKD-MBD guidelines made the following recommendation, which we also recommend:

TABLE 2. Prevalence of HPT in Hemodialysis Patients

	Overall	USA	Canada	Europe
No. patients	12,235	7577	580	4078
Age (y, mean/SE)	63.6 (14.8)	62.7 (14.8)	64.1(14.8)	85.3 (14.6)
Male %	57	55	55	60
Black race	—	36%	—	—
Intact PTH mean/SEM (pg/mL)	456 (402)	467 (411)	516 (415)	427 (379)
Median PTH (pg/mL)/ (median ICR)	342 (277)	350 (275)	385 (364)	322 (269)
PTH >600 pg/mL (%)	19	19	27	17
IV vitamin D use (%)	59	81	8	26
Oral vitamin D use (%)	21	9	58	38
Cinacalcet use (%)	25	28	9	23
Phosphate binder use (%)	86	86	92	85

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Recommendation 3-1: "In patients with CKD G5D* requiring PTH lowering therapy, calcimimetics, calcitriol, or vitamin D analogues, or a combination of calcimimetics with calcitriol or vitamin D analogues should be used for the initial management."[3] (Strong recommendation, moderate-quality evidence.)
*Stage 5 ESKD on chronic dialysis.

The 2017 KDIGO update Work Group made this recommendation noting "there was a lack of uniform consensus among the Work Group members in their interpretation of these data with regard to establishing cinacalcet as the recommended first-line therapy for patients with CKD G5D requiring PTH-lowering therapy. While some felt that only the primary analysis should be used to interpret the outcome, others were equally convinced that the secondary analyses strongly suggested a benefit of treatment with cinacalcet on important patient-level outcomes." These studies have therefore led to physician choice of medical treatment for SHPT. Generally, calcimimetics are more commonly used in patients with higher calcium and/or phosphorus levels, whereas calcitriol or vitamin D analogues are used when calcium and/or phosphorus levels are lower.

Parathyroidectomy for SHPT

There is no absolute PTH value at which PTX is indicated. The 2003 K/DOQI guidelines recommended a sustained level of intact PTH >1000 pg/mL as the level at which PTX should be considered.[69] However, in the 2009 KDIGO guidelines recommend "In patients with CKD stages 3–5D with HPT who fail to respond to medical/pharmacological therapy, we suggest PTX (2B)." The KDIGO guidelines further noted that in addition to the PTH level, patients who are unable to control PTH without precipitating hypercalcemia or hyperphosphatemia should also be considered for PTX, as should patients with anticipated kidney transplant in the ensuing 6 months.[69,70] The availability of a variety of treatments has altered the PTX rates in patients on dialysis in the United States. The observed PTX rates among dialysis patients declined by 30% between 1995 and 1999, with the rate of PTX in 1999 among dialysis patients being 7.16 per 1000 person-years at risk (Fig. 4).[72] A later study of US billing data found that 32,971 parathyroidectomies were performed in patients with ESKD between 2002 and 2011, or 5.4/1000 patients. This decline in PTX rate was also demonstrated in patients with SHPT from 2004 to 2016 using the National Inpatient Sample.[73] Worldwide, there has been a reduction in the number of parathyroidectomies performed in patients undergoing dialysis (Fig. 4).[74] The most recent study examining PTX rates demonstrate that in 379, 835 patients with ESKD on dialysis, the rates of both calcimimetics and PTX for the treatment of SHPT have increased from 2006 to 2016.[75]

HPT treated with PTX is associated with a reduction of PTH, calcium, and phosphorus, which may lead to a reduction in the deleterious effects of PTH toxicity, which include cardiovascular morbidity and renal osteodystrophy. The PTH

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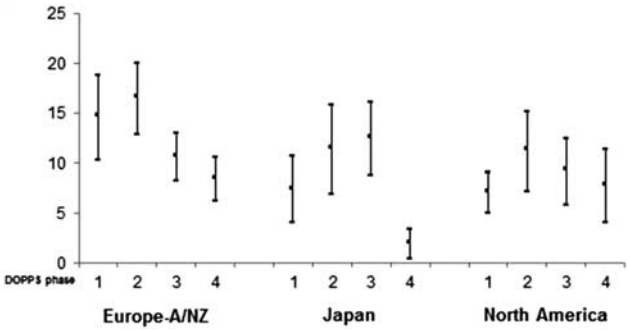

FIGURE 4. Rates of parathyroidectomy between 1996 and 2012. Parathyroidectomy rate (per 1000 patient years). DOPPS phases: 1 = 1996 to 2001, 2 = 2002 to 2004, 3 = 2005 to 2008, 4 = 2009 to 2011, 5 = 2012 to 2015. Eur-A/NZ includes Australia, Belgium, France, Germany, Italy, New Zealand, Spain, Sweden, and the United Kingdom; North America includes Canada and the United States. Adapted from Tentori et al.[71] Adaptations are themselves works protected by copyright. So in order to publish this adaptation, authorization must be obtained both from the owner of the copyright in the original work and from the owner of copyright in the translation or adaptation.

receptor is ubiquitously located, and as demonstrated in previous animal studies, can have direct toxicity at multiple sites in hyperparathyroid state. Massry and Smogorzewski used fifth/sixth nephrectomy rats who underwent a total PTX with repletion of calcium versus PTX with infusion of PTH to clearly demonstrate toxicity on myocardium, pancreas, skeletal muscle, brain, and arteries.[76] In human studies, there is demonstrable improvement of multiple symptoms when comparing uremic patients before and after PTX including, but not limited to pruritis,[77] muscle function,[78] left ventricular function,[79] anemia,[80] health-related quality of life,[81] cognition,[82] and coronary artery calcification.[77–80,82–85] Unfortunately, no randomized trials have compared medical treatment versus PTX.

Studies of long-term outcomes have been published using national databases in multiple countries. In the US-based Medicare United States Renal Data System (USRDS), where patients can be matched for many factors, PTX has been shown to be associated with a 32% lower risk for hip fracture (95% CI: 0.54–0.86, $P = 0.001$) and 31% lower risk for any fractures.[86] Using this same dataset, Kestenbaum et al[87] found an increased early mortality in patients followed for a median of 2.9 year with a relative risk of 2.72 (95% CI: 2.12–3.48, $P < 0.001$) and 1.45 (95% CI: 1.23–1.72, $P < 0.001$) from 0 to 30 and 30 to 90 days, respectively. Thereafter mortality was reduced, reaching 0.85 (95% CI: 0.75–0.97, $P = 0.014$) after 3 years.[87] Costa-Hong et al[88] found in 68 patients with SHPT who were observed matched for comorbidities and PTH level to 50 SHPT patients who underwent a PTX, had an increased cardiovascular risk if they did not undergo a PTX compared with those that did (2.36; 95% CI: 1.11–6.32, $P = 0.02$). Chen at al[89] prospectively followed 161 consecutive PTX compared with 354 hemodialysis patients between 2009 and 2013 and found PTX reduced all-cause and cardiovascular mortality; by multivariate analyses, PTX was associated with a reduction in mortality (HR 0.35; 95% CI: 0.16–0.74). A Swedish Cohort using the Swedish Renal Registry (SRR) identified 423 dialysis patients who had PTX between 1991 and 2009 of 20,056 patients in the registry during the study period, these patients were matched to a total of 1234 patients based on age, sex, cause ESKD (1 case to 5 controls, using PTX date as index date); the patients who underwent PTX had a 0.74 (95% CI: 0.62–0.90) reduction in all-cause mortality.[90] Komaba et al[91] examined patients in the Japanese Society for Dialysis Therapy who underwent PTX to those who had not, they found improved all-cause and cardiovascular mortality in patients who had undergone PTX with a HR of overall mortality of 0.69 (95% CI: 0.58–0.815, $P < 0.001$) and cardiovascular mortality of 0.56 (95% CI: 0.44–0.71, $P < 0.001$), adjusted for age, sex, duration of dialysis, cause of CKD, CRP, dialysate calcium, body mass index, cardiovascular disease, albumin, vitamin D use, calcium, and phosphorus). Figure 5 shows stratified HRs for this propensity matched cohort. Finally, a meta-analysis of 13 studies (n = 22,053) showed an overall benefit of PTX on all-cause mortality of 0.71 (95% CI: 0.61–0.84) adjusted for duration of follow-up.[92]

Taken together, these data suggest that in patients with medically refractory SHPT, PTX is beneficial at reducing cardiovascular and all-cause mortality. The KDIGO guidelines recommend maintaining PTH levels between 2 and 9X normal value. If the PTH exceeds 9X normal value and the PTH has refractory symptoms such as bone and joint pain, fractures, proximal muscle weakness, extraskeletal calcification, calciphylaxis, pruritus, hypercalcemia, or hyperphosphatemia, has been defined as medically refractory SHPT (Table 3). Further investigation with randomized-control trials are needed to truly demonstrate the benefits outlined above are the result of PTX.

Recommendation 3-2: PTX has the potential to reduce cardiovascular and all-cause mortality in patients with SHPT. Patients who fail medical therapy, have complications due to HPT, or prefer surgery should be referred for PTX. (Weak recommendation, moderate-quality evidence.)

The risk of PTX is largely due to perioperative risk related to patient comorbidities and postoperative complications of severe hypocalcemia due to hungry bone syndrome (HBS). Data from the USRDS data set between 2002 and 2011 demonstrated a 3.1% 30-day postoperative mortality rate, however, over a 4-year period, there was a 10% to 15% lower mortality when compared with matched patients not treated with PTX.[87] In the Netherlands, a national database identified 187 cases of PTX in patients with ESKD, with a median age of 46 years.[93] The 30-day mortality and complication rate were 0% and 7.9%, respectively.[93] Over the 25-year follow-up, 7% required re-exploration.[93] Another study using US claims data showed rates for in-hospital mortality decreased from 1.7% to 0.8% between 2002 and 2011 and that the risk of mortality was higher in patients with heart failure and vascular disease.[94] Using the American College of Surgeons National Surgical Quality Improvement Program database between 2006 and 2014, 1661 patients underwent PTX for medically refractory SHPT with a 0.84% overall mortality rate, and 7.9% readmission rate. In the same study 588 patients underwent PTX for THPT with a 0.17% mortality rate, and 5.3% readmission rate.[95] It is also notable that for patients with SHPT and THPT, PTX may also be the most cost-effective strategy when compared with long-term cinacalcet use.[96,97]

Clinical Diagnosis and Management of THPT

THPT is caused by autonomous function of multiple hyperplasic parathyroid gland nodules and is often seen in patients on prolonged dialysis or postkidney transplant. THPT

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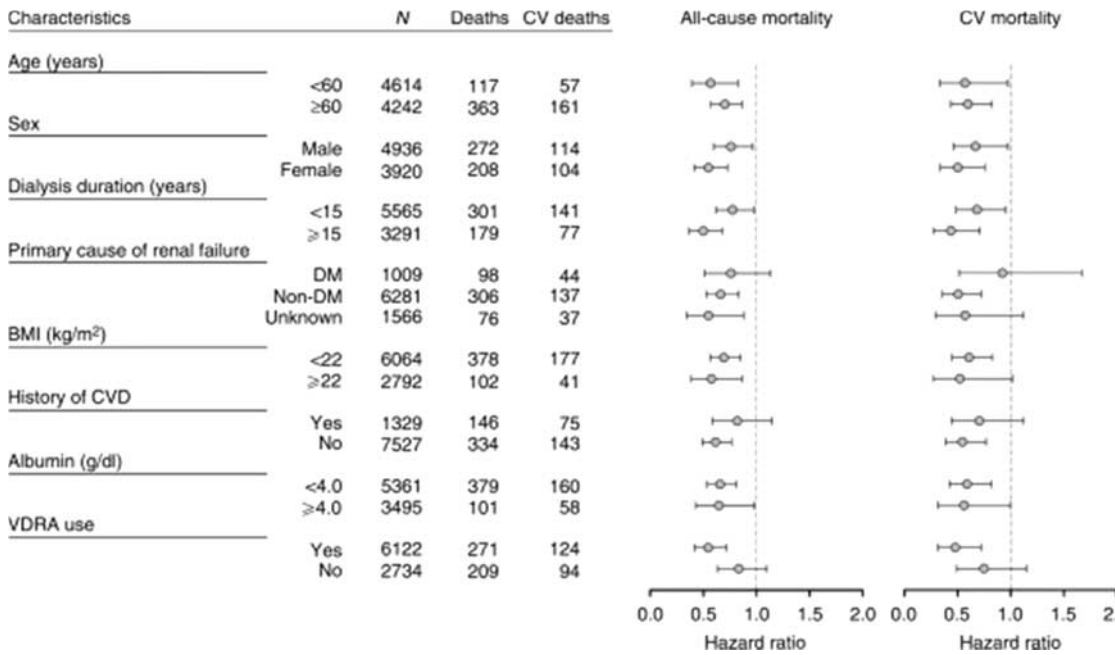

FIGURE 5. Propensity-matched outcomes of Japanese dialysis patients who underwent parathyroidectomy versus those that did not across subgroups. Stratified HRs (and 95% CIs) for all-cause and cardiovascular mortality in the propensity score–matched cohort comparing patients who had undergone PTX with those who had not. BMI indicates body mass index; CV, cardiovascular; CVD, cardiovascular disease; DM, diabetes mellitus; VDRA, vitamin D receptor activator. Adapted from Komaba et al.[91] Adaptations are themselves works protected by copyright. So in order to publish this adaptation, authorization must be obtained both from the owner of the copyright in the original work and from the owner of copyright in the translation or adaptation.

after kidney transplant is defined as a persistently elevated PTH 6 months posttransplantation. It can be characterized by elevated calcium levels in the setting of elevated PTH levels but may also present with normocalcemia. Following kidney transplant, the biochemical changes of CKD-MBD commonly persist, and can make the diagnosis of THPT challengning.[98–101] Decreased vitamin D concentrations are common, with a reported prevalence of up to 81%.[99] The causes are multifactorial, and may include nutritional deficiency, malabsorption, and decreased sun exposure. However, the improvement in kidney function results in an increase in CYP27B1 activity, resulting in increased conversion of calcifediol to calcitriol, which subsequently causes a gradual decrease in PTH concentrations during the first 3 to 6 months after transplantation. However, THPT exists in about 20% to 86% of patients at 12 months.[102–105] Serum calcium concentrations typically drop very early after transplantation, as a result of cessation of therapy with active vitamin D analogues and/or calcium-containing phosphate binders. However, shortly after that initial decrease, calcium concentrations progressively increase, with hypercalcemia developing in a substantial number of patients during the first to third month after transplantation. Hypercalcemia is generally transient, resolving by 6 to 8 months in many patients, but may persist for many years in others. Hypercalcemia is typically associated with HPT and results from enhanced renal tubular reabsorption of calcium under the influence of PTH in the functioning allograft, the effects of calcitriol on the gastrointestinal absorption of calcium, and, potentially, a direct effect of PTH in causing calcium efflux from bone.[98,99] Serum FGF-23 concentrations decrease immediately after transplantation, although they remain inappropriately elevated for the prevailing serum phosphate concentration. By about 3 months after transplantation, serum FGF-23 concentrations are reduced by 89%, and at 12 months FGF-23 concentrations appear to be similar to those in patients without transplantation who have comparable stages of CKD.[99,106] The elevated FGF-23 inhibits CYP27B1 and contributes to the low calcitriol concentrations observed in the early posttransplantation period, likely contributing to the persistent HPT seen after transplantation. Hypophosphatemia is also common during the first 6 months following transplantation, which is likely from the elevated FGF-23 and PTH combined with low calcitriol concentrations.[102,106] Taken together, the changes in the biochemical measures of CKD-MBD fluctuate, and usually improve postkidney transplant by 9 to 12 months. Therefore,

TABLE 3. Potentical Indications for parathyroidectomy in Patients Refractory to Medical Therapy*

Uncontrolled hypercalcemia or hyperphosphatemia
Calciphylaxis or symptomatic severe extraskeletal calcification
Cortical bone fractures
Proximal muscle weakness, bone, or joint pain affecting quality of life
Uncontrolled pruritis resulting in skin lesions and/or affecting quality of life
Patients with imminent kidney transplant who are at risk for severe hypercalcemia post-transplant

*Refractory to medical therapy indicates patients with PTH levels exceeding nine times the upper limit of normal for the assay despite a trial of calcimimetics or other treatments that is either ineffective or stopped due to adverse effects.

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consideration of PTX can be deferred to that time point in the absence of significant hypercalcemia.

Recommendation 3-3: Referral for PTX for THPT should be placed within 12 months postkidney transplant. (Strong recommendation, low-quality evidence.)

There are multiple pharmacologic strategies that are used to treat THPT (Table 4). Treatment recommendations are largely based on strategies used in managing dialysis patients with CKD-MBD. KDIGO published guidelines in 2017 with a special section on the management of CKD-MBD in transplant patients; the recommendations were the subject of a Kidney Disease Outcomes Quality Initiative (KDOQI) commentary the same year.[3,4] Immediately following transplantation, due to the biochemical changes of a newly functioning graft, the guidelines recommended measuring serum calcium and phosphate concentrations weekly, until stable. After stability, the frequency of these measurements should be based on clinical judgment. They suggested that measuring vitamin D concentrations and correction of deficiency should parallel that done in the general population.[3,4]

Cinacalcet has been effective in reducing PTH concentrations to ~50% of baseline in moderate-to-severe THPT.[107,108] In addition to the effective decrease of PTH, cinacalcet could also help control both hypercalcemia and hypophosphatemia. Observations regarding the effects of calcimimetic on kidney allograft function are contradictory. Some groups observed that cinacalcet use was associated with a decrease in glomerular filtration rate (GFR); however, others indicate no difference in kidney function, and no histologic effects on protocol-driven kidney transplant biopsies.[109–112] There is 1 randomized-controlled trial comparing subtotal PTX to cinacalcet for the treatment of hypercalcemia caused by THPT.[113] All patients with subtotal PTX and 67% with cinacalcet achieved normocalcemia. Moreover, the subtotal PTX group had a greater reduction of PTH that was associated with improved bone mineral density in the femoral neck. Vascular calcifications remained unchanged in both groups. A limitation of this study is the small sample size (only 15 participants per arm) and short-term follow-up (12 months).

In addition to the biochemical abnormalities of CKD-MBD, bone fractures posttransplant are of concern. A Cochrane Review of Interventions for preventing and treating bone disease in kidney transplant recipients was published in 2019 and 2020.[114,115] This analysis included 45 studies (2698 subjects) with a median follow-up of 12 months. The analysis focused on fractures and not specifically on correction of HPT and included studies evaluating cinacalcet, denosumab, teriparatide, and PTX. The analysis suggested that bisphosphonates may prevent fractures following kidney transplantation, although the 95% CI included the possibility that bisphosphonate therapy might make little or no difference in fractures. These data are limited in that most of the subjects received bisphosphonates within 1 month of transplantation independent of bone mineral density measurements and were treated for 1 year. Hypocalcemia was observed in some patients treated with bisphosphonates and there was no effect on vascular calcifications or mortality. There was no measurable effect of vitamin D compounds on fractures or vascular calcifications. The Cochrane review concluded there were too few studies with other therapies to draw any conclusions.

In summary, HPT is common among kidney transplant recipients due to the almost ubiquitous development of SHPT and enlarged parathyroid glands during the late-stage CKD and dialysis period. Sequelae of THPT include hypercalcemia, hypophosphatemia, persistently elevated PTH and FGF-23, and increased risk of fractures. Frequently, the use of active vitamin D analogues is limited by hypercalcemia. Treatment with calcimimetics will help control hypercalcemia, but rarely leads to regression of the HPT and does not improve BMD. PTX normalizes hypercalcemia and hypophosphatemia and may have beneficial effects on bone mineral density. However, there are no data on the effects of PTX on reducing fractures in patients with THPT. PTX (subtotal or total) was shown to be more effective at curing THPT compared with those who underwent medical management.[116]

In patients who have received a kidney transplant, preservation of the graft to optimize graft function is of utmost importance. THPT, characterized by hypercalcemia, raises potential for graft dysfunction and even failure.[117–119] PTX for THPT leads to an initial decrease in kidney function.[120–127] Some patients may need to resume dialysis in the year following surgery.[122] However, it can be difficult to differentiate decline due to PTX from the expected decrease in graft function over time.[117,128] Encouragingly, many studies also demonstrate stabilization or improvement in kidney function at longer follow-up durations.[124,126] Total PTX with autotransplantation is associated with poorer kidney function when compared with subtotal PTX or limited resection, which can actually improve GFR.[123,129] Given the potential benefit of PTX, earlier referral for surgery (< 278 days posttransplant) is also associated with improved kidney function postoperatively.[130] Referral to an Endocrine Surgeon for evaluation and follow-up within the 6 months after kidney transplantation in patients that have not had resolution of HPT is necessary to prevent scheduling delays that may affect kidney function if PTX is indicated.

Despite the above noted benefits of PTX, there are reports of no difference in graft survival between patients receiving PTX and patients not receiving PTX. While one retrospective study comparing surgical and nonsurgical THPT reported a 60% decrease in graft survival for the surgical cohort, the study was limited by low patient numbers, and selection bias regarding surgical referrals may have played a role in these outcomes.[120] Other studies comparing surgical groups to posttransplant patients without THPT demonstrate no difference in graft survival overall at 3, 5, and 10 years postoperatively.[121,124,125,131] Cinacalcet is also a treatment option for THPT. When compared with cinacalcet, a randomized control trial demonstrated that PTX leads to improved rates of normocalcemia and

TABLE 4. Pharmacologic Therapies for Tertiary HPT

Nutritional vitamin D	Cholecalciferol	May be useful in high doses
	Ergocalciferol	Cautious with pre-existing hypercalcemia
Vitamin D receptor activators (VDRAs)	Calcitriol	Decrease PTH
	Paricalcitol Doxercalciferol	Increases calcium
Calcimimetics	Cinacalcet	Decreases calcium Cause hypocalcemia Increases phosphate
Antiresorptive agents	Bisphosphonates Calcitonin Denosumab	May decrease calcium May increase PTH

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normalization of PTH levels, as well as improved bone mineral density 1 year posttreatment.[113] Two studies have shown no difference in GFR decline or creatinine at 1 year following surgery or cinacalcet initiation.[113,132] Other retrospective studies concur that cinacalcet normalizes calcium levels but not PTH levels, and that PTX leads to lower rates of graft failure than cinacalcet. Consequently, PTX is the treatment of choice for the preservation of kidney allograft function.[133,134] Notably, additional investigation is needed with randomized-control trials to confirm these benefits of PTX given the potential for confounding factors with these retrospective studies.

Recommendation 3-4: PTX is the treatment of choice for patients with hypercalcemic THPT and therefore should be considered in patients on long-term dialysis or in the postkidney transplant setting. (Strong recommendation, low-quality evidence.)

IMAGING

Parathyroid Imaging

In the setting of PHPT, imaging may be utilized for the localization of a parathyroid adenoma to guide a focused exploration and for the identification of ectopic parathyroid glands.[1] Parathyroid imaging is less accurate in the setting of SHPT and THPT.[135] The recommendation for 4-gland exploration for kidney-related HPT precludes the need for preoperative imaging to localized parathyroid glands in the eutopic locations. Ultrasound imaging to localize ectopic parathyroid glands is often limited, but can be used to rule out concomitant thyroid pathology (see next paragraph).[136–138] A sestamibi scan may be useful in the setting of SHPT for the identification of ectopic mediastinal parathyroid glands if thymectomy is not routinely performed.[139–142]

Concurrent Thyroid Disease

Thyroid nodules are extremely common and occur in over 50% of patients; malignancy is present in 4.0% to 6.5% of thyroid nodules. Hyperthyroidism is present in 1% to 3% of the US population.[143] Many studies have reported the coexistence of thyroid disease and PHPT: thyroid malignancy occurs in 12% to 67% of patients with PHPT, while PHPT is diagnosed in 0.3% to 5.0% of patients with thyroid disease.[144] The prevalence of thyroid disease simultaneous with SHPT is not well reported. Single-institution studies note thyroid carcinoma in 6.5% to 36.0% and nodules or goiter in 51.7% to 64.0% of patients undergoing surgery for SHPT, and thyroid carcinoma in 2.2% of patients receiving surgery for THPT.[145–147] Given the possible high prevalence of concurrent thyroid and parathyroid disease, preoperative evaluation of thyroid disease, including necessary lab and imaging work-up, is often performed. Intraoperative evaluation of thyroid nodules is less sensitive than preoperative ultrasound, and leads to a higher rate of thyroidectomy.[136,148] Preoperative evaluation facilitates identification of thyroid pathology and allows for informed decision-making by the surgeon and patient. All parathyroidectomies for kidney disease require a bilateral exploration, thereby creating a scarred field in the bilateral central necks. Consequently, any need for future thyroid surgery would inherently take place in a reoperative field, which can be associated with higher complication risks. Avoidance of these risks thorough preoperative imaging and thyroidectomy as indicated at the time of PTX is recommended. There are data to suggest that omitting thyroid ultrasound does not harm patients and may prevent unnecessary thyroid interventions.[149] The evaluation for thyroid disease concurrent with kidney-related HPT should be the same as for patients without parathyroid disease. The incident and prognosis of thyroid cancers in the patients undergoing solid-organ transplant is similar to the general population.[150–152] In patients with kidney-related HPT, the presence of thyroid cancer or thyroid nodules should not preclude kidney transplant. Evidence-based guidelines, such as those published by the American Association of Endocrine Surgeons or the American Thyroid Association, should be followed for the management of concurrent thyroid disease.[144,153] If a thyroid nodule meets criteria for biopsy via fine-needle aspiration, the aspirate should be sent for PTH levels or molecular profiling that includes classifiers for parathyroid tissue as the nodule may be an intrathyroidal parathyroid gland.[1] Cytology on an intrathyroidal parathyroid gland may be misread as a follicular neoplasm.[1]

Recommendation 4-1: Cervical ultrasound may be used to assess for concomitant thyroid pathology. (Weak recommendation, low-quality evidence.)

Recommendation 4-2: Preoperative imaging is limited for the localization of parathyroid glands in the expected anatomic locations but may be helpful for parathyroid glands in ectopic locations. (Strong recommendation, low-quality evidence.)

PREOPERATIVE AND PERIOPERATIVE CARE

Universal/Standard Consent and Counseling

Five basic elements comprise informed consent provided by a patient: voluntarism, capacity, disclosure, understanding, and ultimate decision-making.[154] For patients' consent to be valid, patients must be given information about a procedure's risks, benefits, and alternatives, and offered the opportunity to have questions answered before deciding whether to undergo the procedure. This discussion can also be helpful to manage patients' expectations about success of and recovery from an operation, including improving their ability to understand any complications that may arise.[155] Many factors influence patients' and caregivers' understanding and retention of information provided during the informed consent process.[156] Several studies have identified that a significant number of patients do not gain a comprehensive picture of their surgical journey by merely following the steps necessary to sign a surgical consent form.[157] Data from other types of procedures suggest that use of visual aids such as pamphlets, diagrams, and multimedia or web-based platforms can enhance perioperative patient understanding.[158]

When counseling patients about surgery, not all members of the team are able to provide a sufficiently individualized risk-benefit profile to patients and their caregivers, or to correctly answer their questions.[159] Best practice is for the patient, and caregiver if present, to be informed by the operating surgeon of the major risks of the operation, including those specific to PTX, surgery in general, and patient-specific risks. Patients should also be informed about recommendations for postoperative pain

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management techniques and safe opioid use. Finally, the consent process should be clearly documented in the patient's chart.

Interdisciplinary Communication

Communication lapses among healthcare professionals are one of the most common causes of sentinel events negatively impacting patient safety.[160] There remains no universally accepted model for transmission of perioperative information. A recent study designed checklists for different stages of the care process in thyroidectomy, including preoperative evaluation, intraoperative findings, and postoperative data, events, and plans.[160] Synoptic operative reports have also been demonstrated to facilitate systematic documentation. Mutual multidisciplinary communication is an important component of optimal patient care and should be systematically encouraged.

Risk Stratification

No guidelines specifically address preoperative evaluation for individuals with advanced CKD. Patient with CKD represent a high-risk population with high rates of concomitant atherosclerotic and calcific coronary artery disease, calcific valvular disease, peripheral arterial disease, and heart failure.[161] Patients with CKD have high rates of adverse perioperative outcomes including acute kidney injury (AKI), cardiovascular events, and mortality.[162] Optimization in the preoperative period should include special risk assessment of cardiovascular risk, coagulopathy, management of hypertension, management of immunosuppression, and timing of dialysis, if needed. One study that examined outcomes of elective surgery in patients above 65 years of age undergoing chronic dialysis identified a 30-day mortality rate of 12.7% in this group versus 1.5% in those without CKD, however, it should be noted that head and neck procedures comprised <6% of patients in this study.[163] A recent systematic review of perioperative optimization of patients with ESKD and posttransplant patients recommends assessment of risk for AKI, cardiopulmonary events, and management of hypertension, anemia and dialysis. Patients with ESKD or who have received a kidney transplant frequently have a number of comorbid conditions.[164] Collectively, these conditions increase the risk of intraoperative or postoperative cardiac events or other complications.[164,165] Preoperative risk assessment and optimization are necessary to mitigate these risks.[166]

AKI Risk Assessment and Dialysis Management

AKI has been examined in patients undergoing PTX with several studies suggesting that makers of kidney function (eg, GFR and creatinine levels) acutely worsen in the immediate postoperative period.[121,167–170] Although both serum creatinine and eGFR are strong predictors of postoperative AKI, risk prediction models that use eGFR demonstrate superior predictive power, thus it is recommended that baseline kidney function using eGFR be assessed within 1 month of surgery, and prediction tools such as the General Surgery AKI Risk Index can be utilized to determine the risk of AKI perioperatively.[171–173]

Recommendation 5-1: Patients with CKD who are not dialysis dependent undergoing PTX should have baseline eGFR assessed at least 1 month before surgery to determine risk for perioperative acute kidney injury. (Strong recommendation, low-quality evidence.)

Surgery should be scheduled to avoid too long an intradialytic period, as data from the DOPPS cohort showed increased all-cause mortality (HR: 1.40) in the longest interval without dialysis for all dialysis patients undergoing an operation.[174] Additional studies have shown increased risk for sudden cardiac death and arrhythmias in prolonged interdialytic periods.[175,176] Potassium derangements can contribute to arrhythmias and major adverse cardiac events, and hyperkalemia with a potassium of 6.0 mmol/L has been described in 10% of maintenance dialysis patients, even if adequate dialysis is being performed.[177–180] Hyperkalemia should be avoided by ensuring recent dialysis and evaluating perioperative electrolytes, typically on the day of surgery.[181]

Patients on dialysis should be dialyzed the day before surgery to optimize volume status and metabolic parameters. This may require adjustment of the patient's regular hemodialysis schedule. Inpatient nephrologists should be alerted in advance of the PTX to coordinate dialysis postoperatively. Coordination of care is paramount due to the potential for life-threatening postoperative hypocalcemia.

Recommendation 5-2: Patients on maintenance hemodialysis undergoing PTX should have electrolytes assessed as close to surgery as possible. (Strong recommendation, moderate-quality evidence.)

Recommendation 5-3: Dialysis routines should be maintained in the perioperative setting. Consider scheduling dialysis to occur within 24 hours before surgery. (Strong recommendation, moderate-quality evidence.)

Cardiopulmonary Risk Assessment

CKD, ESKD, and kidney transplantation are independent risk factors for adverse cardiovascular events.[182] Clinical risk for major adverse cardiac events include unstable coronary syndrome, decompensated heart failure, significant arrhythmia, and severe valvular heart disease.[182] If these risks are not present, risk stratification should be performed based on functional capacity using metabolic equivalents criteria.[182] Because CKD is considered to be a risk factor for coronary artery disease in the intermediate cardiac risk population, preoperative electrocardiogram and echocardiogram are recommended for all patients with ESKD or postkidney transplant.[182,183] For patients on maintenance dialysis, noninvasive stress testing should also be considered.[183] Any patient with additional cardiac risk factors should undergo preoperative evaluation and optimization by a cardiologist.[184]

Recommendation 5-4: Preoperative cardiac risk stratification should include identification of clinical risk factors for major adverse cardiac events and assessment of functional capacity. (Strong recommendation, moderate-quality evidence.)

Recommendation 5-5: Baseline preoperative electrocardiogram and echocardiogram should be obtained in all patients with CKD, ESKD, and kidney transplant recipients with diminished functional capacity. (Strong recommendation, moderate-quality evidence.)

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Hypertension Management

Hypertension is associated with increased risk for adverse outcomes in a perioperative setting, including AKI, cardiovascular outcomes, and end-organ damage, and is particularly important in the CKD and posttransplant population due to risk for AKI, CKD progression, and loss of residual kidney function.[185] One study found that severely uncontrolled hypertension (BP >180/110 mm Hg) was associated with increased risk for adverse cardiovascular and kidney outcomes, though these data were not specific for a CKD or post-transplant population.[186] The ideal target for blood pressure control is <140/90, but abrupt reduction in blood pressure to achieve this figure may confer additional risk due to impaired kidney and cerebral autoregulation.[187] Ongoing hypertension management can include continuation of chronic antihypertensive therapies up to the morning of surgery, though there is insufficient evidence to definitively recommend either continuation or withdrawal of renin-angiotensin-aldosterone system blockade agents, but risk for hyperkalemia and hemodynamic instability with anesthetic exposure should be considered if these medications are continued.[188,189]

Recommendation 5-6: Blood pressure should be controlled perioperatively and maintained at a level <180/110 mm Hg perioperatively. Patients should continue their existing antihypertensive regimen if well-controlled, with no evidence to recommend for or against discontinuation of renin-angiotensin-aldosterone system blockade. (Strong recommendation, low-quality evidence.)

Anemia and Bleeding Management

Patients with ESKD or on hemodialysis are at increased risk of perioperative bleeding, which can exacerbate both anemia from iron and relative erythropoietin deficiency, and platelet dysfunction due to uremia.[190,191] To prevent complications due to postoperative anemia, hemoglobin should be optimized to the KDIGO target of 10 to 11 g/dL which can be accomplished with erythropoietin stimulating agents and intravenous iron, as needed.[192] In addition, adequate, recent dialysis also improves platelet function abnormalities and can reduce bleeding risk, though there is insufficient evidence as to whether increasing preoperative dialysis dose reduces bleeding events or has an impact on surgical outcomes.[193] Surgeons should also discuss the risks of transfusion and implications of sensitization for patients who are eligible for kidney transplantation. A single dose of desmopressin at 0.04 μg/kg can be administered to patients with a history of excessive bleeding, or those in whom excessive bleeding is identified intraoperatively.[194] Any anticoagulants should be held based on the specific medication's elimination half-life, patient-specific bleeding risk, and procedural risk, accounting for the patient's eGFR.[177,195]

Recommendation 5-7: Patients on maintenance hemodialysis should have hemoglobin concentrations optimized preoperatively. (Weak recommendation, low-quality evidence.)

Recommendation 5-8: A long interdialytic interval should be avoided before surgery to optimize platelet function, and desmopressin can be administered intraoperatively to treat excessive bleeding due to platelet dysfunction. (Strong recommendation, moderate-quality evidence.)

Vocal Cord Evaluation

During the preoperative evaluation, patients should be asked about voice changes as well as a history of operations that place the recurrent laryngeal or vagus nerves at risk (thyroidectomy, carotid endarterectomy, cervical spine operations with an anterior approach, etc.). If the patient's history is notable for one of these operations or a persistent voice change, the vocal cords should be evaluated. Options include a lighted laryngeal mirror examination, flexible laryngoscopy, videostrobolaryngoscopy, and transcutaneous laryngeal ultrasonography/vocal cord ultrasonography.[1]

Recommendation 5-9: Patients with a voice change or a surgical history placing the recurrent laryngeal or vagus nerves at risk should receive a preoperative vocal cord assessment. (Strong recommendation, weak evidence.)

Vitamin D

Vitamin D deficiency is very common and is one etiology for SHPT, independent of kidney function. Clinical practice guidelines recommend that vitamin D supplementation in the form of ergocalciferol (vitamin D_2) or cholecalciferol (vitamin D_3) be initiated in patients with SHPT who are vitamin D deficient, although definitions for vitamin D deficiency, the optimal 25-hydroxyvitamin D concentration to target, and vitamin D repletion regimens vary.[196] In patients with CKD who also manifest deficient 25-hydroxyvitamin D stores, use of vitamin D_2 or vitamin D_3 has been advocated to lower PTH, but the results are mixed.[44,197] In patients on dialysis with functionally anephric kidneys such that the 25(OH)D cannot be converted to $1,25(OH)_2D_3$, supplementation may not be needed.

Preoperative treatment with oral calcium and calcitriol can prevent or mitigate the development of HBS postoperatively.[198] Calcitriol is frequently used in patients with ESKD to mitigate the degree of HPT, regardless of a plan for surgery.[2] If patients are not already on calcitriol, initiation of calcitriol for at least 2 days preoperatively is associated with less severe hypocalcemia and reduces postoperative length of stay and need for intravenous calcium replacement.[166,199,200] There is some evidence from a single-institution study to support the preoperative administration of bisphosphonates during hemodialysis in the days before surgery to prevent symptomatic postoperative hypocalcemia.[201]

Recommendation 5-10: Patients with SHPT and vitamin D deficiency should have vitamin D repletion before PTX. (Strong recommendation in CKD and postkidney transplant; weak recommendation in ESKD, low-quality evidence.)

Recommendation 5-11: Calcitriol should be initiated at least 2 days before PTX. (Strong recommendation, moderate-quality evidence.)

Calcium

The recommended amount of daily elemental calcium intake in adults is 1000 to 1200 mg and appears appropriate for patients with SHPT or THPT as well.[196] This amount should

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not be exceeded in CKD and ESKD patients with SHPT or THPT as kidney calcium excretion is impaired.[44] In this setting, excess calcium loading can lead to ectopic calcification, including in large conduit arteries and the coronary arteries, in tendons and ligaments, and in other soft tissues. These forms of calcification have been associated with increased risks of cardiovascular events, musculoskeletal injuries, and mortality.[202,203]

Patients who have undergone prior gastric bypass surgery should be counseled about a higher risk of severe postoperative hypocalcemia. Roux-en-Y gastric bypass routes the flow of nutrients away from most of the stomach, duodenum, and proximal jejunum, thereby limiting much ability to absorb calcium and vitamin D, with calcium citrate being the formulation that is best absorbed.[204] The incidence of refractory hypocalcemia following PTX in the gastric bypass patient population is likely underreported.[205] In addition, a typical-sized normal parathyroid remnant (40–60 mg) may not be sufficient to maintain normocalcemia, which has been proposed as a reason to leave at least 1 entire parathyroid gland intact during PTX.[206]

Recommendation 5-12: Most patients with SHPT or THPT should follow national guidelines for calcium intake. (Strong recommendation, moderate-quality evidence.)

Recommendation 5-13: Gastric bypass patients should be counseled about the higher risk of severe postoperative hypocalcemia after multigland PTX. (Strong recommendation, low-quality evidence.)

Calcimimetics

Patients on calcimimetics preoperatively may have increased resected parathyroid gland weight and higher PTH levels.[207] In addition, patients on calcimimetics preoperatively have increased rates of bone turnover and HBS.[109,208,209] It is unclear if these effects are the direct result of calcimimetic use or a reflection of more severe HPT. Despite these potential negative effects for patients undergoing PTX, calcimimetics have not been demonstrated to adversely affect the rate of surgical cure.[210–213] No data exist to guide the optimal timing of calcimimetic discontinuation before PTX, with practice variation in stopping cinacalcet anywhere from 14 days before surgery to the morning of surgery. Furthermore, all currently published data on calcimimetics in the perioperative period of PTX focused on cinacalcet. Recently, etelcalcetide, an intravenous calcimimetic that is longer acting than cinacalcet, has been approved for use in patients with ESKD and SHPT.[58] Additional research is needed to determine the optimal timing of PTX in patients being treated with etelcalcetide.

While calcimimetics decrease serum calcium and PTH levels, enthusiasm for them is tempered by their cost and side effects. In one study of patients referred to surgery for THPT, inability to afford the medication was a contributing factor in 26%.[130] In addition, significant percentages of patients report intolerance of the medication due to nausea.[130] Authors have additionally reported that use of calcimimetics results in decreased kidney function among kidney transplant recipients, but this appears to be transient and has been shown to reverse following cessation of the medication.[109,119,214,215] Use of calcimimetics in SHPT or THPT can help to lessen the negative effects of elevated PTH without adversely affecting the rate of surgical cure.

Recommendation 5-14: Use of calcimimetics can help to lessen the negative effects of elevated PTH without adversely affecting the rate of surgical cure in patients with pretransplant ESKD SHPT and THPT and may be considered in these patients. (Weak recommendation, low-quality evidence.)

Antibiotics

Surgical site infection is an unusual event after standard transcervical PTX, with a reported incidence of 0.09% to 2%.[216] Antimicrobial prophylaxis (AMP) for clean (uninfected wounds without inflammation and the respiratory, alimentary, genitourinary tracts are not entered) operations in the head and neck was not recommended in a 2013 guideline from the American Society of Health-System Pharmacists, with moderate-quality evidence.[217] The safety of surgery without AMP in clean thyroid and parathyroid procedures was recently confirmed in a single-institution randomized trial.[216] Routine AMP is still commonly used for clean thyroid and parathyroid operations in Japan and other countries.[218,219] Patients with SHPT and THPT may be more likely to present with patient-specific factors such as the presence of a tunneled dialysis catheter, immunosuppression, diabetes, or obesity, that can increase the risk of surgical site infection, or have potentially longer operative time with four-gland exploration. Such patients deemed to be high-risk may benefit from selective use of AMP, which should consist of gram-positive coverage administered before or on induction.[220] If utilized, the dose of AMP should be adjusted based on kidney function in all patients. With rare exceptions, postoperative AMP is not indicated.

Recommendation 5-15: AMP is not indicated in most cases of standard transcervical parathyroid surgery, but selective use should be considered in patients with factors placing them at higher risk for surgical site infection. (Strong recommendation, moderate-quality evidence.)

Steroids

In patients who do not routinely take glucocorticoids, a reduction in postoperative nausea and vomiting as well as postoperative pain in thyroidectomy patients was demonstrated with use of intravenous dexamethasone in both a meta-analysis and randomized controlled trial.[221,222] Efficacy was enhanced with higher doses of dexamethasone (8–10 mg) and when given before induction of anesthesia, as the onset of action is 1 to 2 hours following administration.[223] In addition, the use of perioperative or intraoperative steroids may reduce the rate of voice dysfunction and recovery time of recurrent laryngeal nerve palsy.[221,224–229]

Recommendation 5-16: Before PTX, in the absence of contraindications, a single preoperative dose of dexamethasone should be considered to reduce nausea, vomiting, and pain. (Strong recommendation, high-quality evidence.)

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Patients with SHPT and THPT often take glucocorticoids as part of a postkidney transplant immunosuppression regimen. Perioperative management of steroids has been an area of controversy and ongoing debate, with few high-quality studies to guide clinical practice in this subset of patients. Recent data suggest that stress-dose steroids may not be necessary, even in patients with confirmed hypothalamic-pituitary-adrenal (HPA) axis suppression.[230] Instead, a patient's typical dose of steroids can be maintained perioperatively, including administration on the day of surgery, along with vigilance for signs of symptoms of adrenal insufficiency that, if identified, should be treated with rescue dose steroids.[231–233] Patients on low-dose chronic steroids (equivalent to 5 mg of prednisone per day or less) do not need stress-dose steroid administration, while patients on high-dose steroids (equivalent of prednisone 20 mg/d for at least 3 weeks) do require stress-dose administration.[234] Patients in either of these groups do not need preoperative evaluation of their HPA axis unless there is reason to believe it might affect perioperative management, with studies suggesting that this testing does not reliably predict patients that will develop adrenal crisis.[233] Preoperative evaluation of the HPA axis may be helpful for patients on chronic steroids that do not fall into either of the above categories, as stress-dose steroids can be safely withheld with proof of a functioning HPA axis. An additional approach to management of the patient on chronic steroids is to assess the anticipated surgical stress to determine the appropriate perioperative steroid dose.[234] Patients that do require stress-dose or rescue steroids should be treated with hydrocortisone as the drug of choice.[235,236]

Chronic steroid therapy is also associated with impaired wound healing, hyperglycemia, and immunosuppression. Perioperative stress-dose steroids have historically thought to increase these risks, but this has recently been challenged.[237,238] No randomized controlled trials exist to assess the effect of perioperative glucocorticoid administration on postsurgical complications in patients undergoing PTX. A retrospective study of kidney transplant patients undergoing surgery to treat lymphocele demonstrated that perioperative stress-dose steroids resulted in hyperglycemia, but did not demonstrate statistically significant differences in adrenal insufficiency or wound-related complications compared with patients who were not treated with stress-dose steroids.[239] The authors concluded that stress-dose steroids increased the risk for perioperative hyperglycemia without apparent clinical benefit.[239]

Recommendation 5-17: Patients on chronic steroid therapy should be stratified by risk of HPA axis suppression and receive preoperative HPA axis testing and perioperative stress dose steroids accordingly. (Strong recommendation, low-quality evidence.)

Venous Thromboembolism (VTE) Prophylaxis

Although VTE prophylaxis has become relatively routine in longer surgical procedures, and those that are more physiologically extensive or followed by bedrest, it remains controversial for routine PTX. The risk for VTE after thyroid and parathyroid surgery is quite low (0.02%–0.2%), and the risk of postoperative hemorrhage can be significant, potentially leading to acute airway compromise requiring emergency decompression.[166,240–242] This leads many parathyroid surgeons to believe that the risk of bleeding outweighs the potential benefits of VTE prophylaxis, particularly given that VTE occurred following only 0.16% of thyroid and parathyroid operations in a large National Surgical Quality Improvement Program (NSQIP) study, and the authors deemed that VTE prophylaxis should be reserved for those patients determined to be at high risk.[166,240] Mechanical thromboprophylaxis with intermittent pneumatic compression devices and immediate postoperative ambulation should be utilized routinely after PTX.

Recommendation 5-18: Chemical venous thromboembolism prophylaxis for PTX should be reserved for patients determined to be at high risk for venous thromboembolism. (Strong recommendation, low-quality evidence.)

Pain Control

The United States Centers for Disease Control and Prevention (CDC) guidelines on the use of opioids for chronic pain highlight a preference for non-opioid and non-pharmacologic therapy, and caution that opioid dependence can stem from prescriptions given for acute pain control.[243] Studies have demonstrated that 6% of patients overall persist in opioid use 90 days after surgery whether they underwent a major or minor surgical procedure.[244] Opioid consumption has also been correlated with the amount prescribed with an average of 0.53 more pills consumed for every additional pill prescribed.[245,246]

Recent studies of interventions designed to decrease opioid use in thyroid and parathyroid surgery by increasing patient education about use of non-opioid adjuncts have met with success. In one study of over 1700 patients in which these strategies were utilized, 57% of patients undergoing parathyroid surgery were discharged without any opioids.[247] If opioids are prescribed, the lowest effective, immediate-release options are preferred. For many patients, postoperative pain after PTX can be effectively managed with opioid-free regimens, if not with low-dose oral morphine equivalent prescriptions.[248–252]

As nonopioid pain management strategies have become more common, most regimens include nonsteroidal anti-inflammatory drugs (NSAIDs). NSAIDs are associated with higher rates of gastrointestinal toxicity in patients with ESKD and are generally contraindicated in CKD patients and post-transplant patients, as the combination with calcineurin inhibitors can precipitate vasoconstriction and reduced kidney function.[253] Use of acetaminophen as a centrally acting non-opioid agent is well tolerated in patients with normal hepatic function.

Recommendation 5-19: Use of nonopioid and nonpharmacologic therapies combined with patient education should be first-line pain management after PTX. (Strong recommendation, moderate-quality evidence.)

Recommendation 5-20: If opioids are prescribed for postoperative pain management, the lowest effective dose of immediate release opioids (< 10 oral morphine equivalents) should be prescribed. (Strong recommendation, moderate-quality evidence.)

Recommendation 5-21: Caution should be used with oral and parenteral NSAIDs administration in patients with CKD, ESKD, or postkidney transplant to avoid risks of GI toxicity and decreased kidney function. (Strong recommendation, moderate-quality evidence.)

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PLANNING OF SURGERY

Goal of Surgery

The goal of PTX for kidney-related HPT is to decrease the source of the excess PTH. This must be balanced with the potential for permanent postoperative hypoparathyroidism and recurrent or persistent HPT.

In patients with prekidney transplant HPT, the potential for future kidney transplantation should be incorporated into the surgical strategy. These patients will depend on PTH to maintain serum calcium levels, especially in the predialysis and posttransplant settings.[69] Consequently, total PTX without autotransplantation should not be performed in patients awaiting a kidney transplant due to high risk for hypoparathyroidism.[69,254] Subtotal PTX is the optimal choice for patients with a high likelihood of kidney transplant, while patients unlikely to receive a transplant may benefit from total PTX without autotransplantation.[255] Patients with HPT postkidney transplant require management of associated hypercalcemia and elevated PTH levels to prevent subsequent allograft dysfunction.[109,119,214,215]

Recommendation 6-1: The patient's dialysis status and likelihood of obtaining a kidney transplant should be considered when planning for subtotal PTX, total PTX with autotransplant, or total PTX without autotransplant. (Strong recommendation, low-quality evidence.)

Positioning, Incision, Exposure, and Closure

Either general anesthesia or locoregional analgesia (cervical block) with intravenous sedation are employed.[1,166] While local anesthesia can be performed safely in this patient population, general anesthesia is more commonly utilized because of the need for bilateral exploration and patient comorbidities.[256] The patient's arms should be tucked at the side to facilitate access to the operative field. If an autotransplantation is planned, the graft site of choice (see below) can be left out and prepped into the field at this time, or the graft site can be exposed and prepped when the autotransplant is ready.[166] The patient's neck is then extended and the patient placed into the supine or semi-Fowler position.

After skin preparation, an incision is made in a skin crease or Langer's line parallel to other skin creases to assist in attaining a cosmetic result. The incision is ideally placed either 1 to 2 fingerbreadths inferior to the cricoid cartilage, or 1 to 2 fingerbreadths superior to the sternal notch.[166] If no optimal skin crease is available, or if a surgeon must choose between 2 skin creases in this region, a lower incision is generally preferred to facilitate access to the upper mediastinum and thymus. During PTX, the recurrent laryngeal nerve (RLN) can be identified, and the path of the nerve used to guide exploration for inferior or superior parathyroid glands; due to large gland size and auto-infarction, the nerve may be adherent to and intimately involved with superior glands, so great care is needed to avoid injury.

All 4 parathyroid glands should be identified before resection. This is a key point, since it is not always evident which is the most normal-appearing gland until all 4 glands are identified for comparison. Once identified, the extent of PTX is determined as described below. The glands should be carefully handled to avoid inadvertent devascularization or disruption of the capsule. The vascular pedicle of each gland to be resected is isolated, clipped, and divided. At the conclusion of the operation, the operative field should be thoroughly examined for hemostasis. A Valsalva maneuver with the patient in Trendelenburg position mimics a cough and can help identify bleeding vessels. Hemostatic agents can be placed in the surgical beds and over the internal jugular vein blood draw site if used to for ioPTH monitoring. Drains are not routinely used.[166] The neck is then closed in a layered manner with a small gap in the strap muscles at the inferior aspect to allow for drainage from the deep to superficial neck.

Recommendation 6-2: In patients undergoing an initial PTX for SHPT and THPT, all 4 parathyroid glands should be identified and assessed before surgical resection. Parathyroid glands should be handled carefully to avoid capsule disruption. (Strong recommendation, low-quality evidence.)

SHPT: Operative Principles

The goal of surgery for SHPT is to reduce the overproduction of PTH via resection of hyperplastic and/or neoplastic parathyroid glands. The risks of postoperative hypocalcemia and hypoparathyroidism must be weighed against the risks of persistent or recurrent HPT, particularly in the setting of ongoing ESKD, which will continue to stimulate PTH production.[257–259] There are 3 operative approaches commonly used for kidney-related HPT. No one operative approach has been uniformly recommended. Consequently, the choice of operation lies with the surgeon.[260] Surgeons may perform just one type of operation, or may select the operation based upon patient characteristics such as age and comorbidities. The likelihood of receiving a kidney transplant, thereby restoring normal kidney function, is one important characteristic: patients likely to receive a kidney transplant may be more suited to operations that preserve a remnant parathyroid gland, while patients unlikely to receive a transplant may benefit from complete resection.[261,262] The 3 operations are described in detail below, followed by a comparison of the risk of recurrence and hypoparathyroidism between the 3 techniques.

Subtotal PTX

Subtotal PTX is the resection of majority of the parathyroid glands. First described by Stanbury and colleagues in 1960 as the removal of 3.5 glands, this operation preserves a remnant of parathyroid tissue with its original blood supply intact, ideally lowering the severity and duration of postoperative hypocalcemia.[215,257,263] Bilateral transcervical thymectomy is often performed in order to remove supernumerary glands or parathyroid rests (see section on thymectomy below). In this operation, all 4 glands are first identified and examined.[264–266] Frozen section can be performed to confirm parathyroid tissue but is not necessary for experienced surgeons. Once all 4 glands are located, one gland is selected to remain in situ. The most "normal"-appearing gland (ideally without nodules representing autonomously functioning clones of transformed parathyroid cells) or the smallest gland with a good vascular pedicle is chosen.[264,266–270] This gland is partially resected sharply and the remnant is marked with a non-absorbable suture and/or metal clip in the event future identification is needed at re-exploration.[264,265,270] There is no consensus on the size of the remnant left behind; various studies have described remnants weighing 30 to 50 mg, 40 to 50 mg or 40 to 80 mg, or measuring 1/3 to 1/2 the gland, half the size of a normal gland, 2 to 3 times the size of a normal gland, or a 3 × 3 × 3 mm remnant.[264,265,267–269,271,272] Once the remnant is marked and examined for viability, the remaining glands can be

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excised.[166] Upon closing, some surgeons mark the level of the remnant with a non-absorbable suture in the strap muscles.

If reoperation for recurrence is needed, there is increased potential for RLN injury due to the scarred operative field.[215] For this reason, some surgeons prefer to leave behind an inferior parathyroid gland if possible, as these are located more anteriorly and may be more accessible for resection.[166,199,269] However, inferior glands may have a more tenuous blood supply, especially if cervical thymectomy is performed. Superior gland remnants may be tacked to the thyroid capsule in a tension-free manner to retract them away from the RLN. If at any point during the operation the remnant gland appears nonviable, the surgeon should resect that gland and leave a different remnant gland in place or perform total PTX with autotransplantation.[199]

Recommendation 6-3: In SHPT, subtotal PTX should leave behind a well-vascularized remnant of the most normal-appearing gland. (Strong recommendation, low-quality evidence.)

Total PTX Without Autotransplantation

A total PTX, first described by Ogg[273] in 1967, removes all identified parathyroid glands. No gland remnants are left behind, and no pieces of parathyroid tissue are reimplanted. The goal of removing all parathyroid glands is to eliminate the risk of persistent or recurrent disease. Consequently, some have advocated performing this procedure in patients with a long-life expectancy on permanent dialysis.[267,274] This procedure should not be performed on patients who have potential to receive a kidney transplant. Following total PTX, patients rely on residual parathyroid tissue in the form of supernumerary glands or parathyroid rests. The majority of patients receiving total PTX have detectable PTH levels in the immediate postoperative period.[275–277] Of the patients initially hypoparathyroid postoperatively, the majority develop measurable PTH levels months to years later.[261,275,278] The potential for hypocalcemia stemming from postoperative hypoparathyroidism is concerning, although rates of transient hypocalcemia requiring supplementation range from 0 to 76%.[276,277,279] Permanent hypocalcemia is reported in just one study, in 5 out of 20 patients (25%).[280]

Recommendation 6-4: Total PTX without autotransplantation should not be performed on patients with CKD who may undergo kidney transplantation. (Strong recommendation, moderate-quality evidence.)

Total PTX With Autotransplantation

Total PTX with autotransplantation removes all parathyroid tissue in the neck while providing an alternate source of PTH. Transcervical thymectomy is often performed as well. Many surgeons perform total PTX with autotransplantation as their procedure of choice.[281] Others perform total PTX with autotransplantation only if there is a high desire to avoid reoperative neck surgery,[24] or in certain clinical situations, such as concern for the viability of the parathyroid remnant based on the absence of a well-defined vascular pedicle, if only three parathyroid glands are identified,[264] if the ioPTH level meets a certain threshold, or if there is a functioning kidney transplant.[281] Total PTX with autotransplantation is initially carried out like a total PTX: all 4 glands are identified and then resected.[265] Once this is complete, a parathyroid autograft is prepared and a graft site is chosen. Confirmation of parathyroid tissue via frozen section is recommended to prevent inadvertent autotransplantation of lymphoid, adipose or thyroid tissue.[282] In addition, real-time histologic examination of parathyroid tissue can help to select the optimal specimen. Diffuse hyperplastic tissue is associated with lower long-term PTH levels, so selection of diffuse hyperplastic tissue for reimplantation rather than nodular tissue after identification on frozen section has been advocated by some.[283,284] Alternatively, a stereo magnifier can be used to identify tissue with stromal fat cells that have lower proliferative potential.[262,285] Many surgeons simply choose the most normal-appearing or smallest gland.[266,286]

Various amounts of tissue to implant have been reported in the literature. In general, 5 to 30 fragments measuring $1 \times 1 \times 1$ mm to $1 \times 1 \times 3$ mm are transplanted, and the graft site is closed with a permanent suture that also serves to mark the site.[199,259,260,265,270,284,286–294] Some surgeons create multiple "pockets" underneath the skin incision and transplant one or more fragments into each.[271,284,294–296] A number of graft sites have been also reported, the most common being the nondominant or nonshunt brachioradialis.[166,259,265,266,270,287,290–294,297,298] The principle advantage of forearm autografting is the ability to define graft-dependent recurrent disease by demonstrating a clear PTH gradient between blood samples taken from the graft-bearing and ungrafted arms. In addition, if graft-dependent recurrent disease is demonstrated, the option of complete removal of the grafted brachioradialis exists without subsequent functional deficit. If the nondominant forearm is chosen, there should be a preoperative discussion about the patient's handedness and current and future dialysis access plans; care should be taken to avoid injuring a dialysis graft or fistula intraoperatively.[166] The sternocleidomastoid is the second most common graft site, with the benefit of requiring only one operative incision.[132,281,286,288,289,299] Other implantation sites, including an intramuscular or subcutaneous presternal site, muscles of the thigh anterior tibialis or trapezius, subcutaneous tissues of the nondominant forearm, and the abdominal adipose tissue are less frequently used.[24,132,290,300–306] Importantly, the autograft site chosen does not impact the risk of postoperative bleeding or voice changes, and the duration of surgery is not significantly different from a total PTX alone.[286,290] Creation of an autograft has additional considerations. Following autotransplantation, there is a delay in parathyroid function as the autotransplanted parathyroid tissue needs time to recruit blood supply. There is the potential for late graft failure, possibly due to scar maturation causing local ischemia.[272,297] Conversely, the grafts can become hyperplastic with time and can infiltrate into surrounding structures and blood vessels.[272,307] In addition, THPT can develop at the site of an autograft after kidney transplantation due to the persistence of nodular tissue in the autograft.[308] If resection due to recurrence or tertiary disease is indicated, wide excision of the surrounding tissue is indicated. Multiple surgeries may be required for complete resection, leading some surgeons to use surgical approaches without an autograft, or to implant into subcutaneous tissues instead.[305,307]

Recommendation 6-5: For autotransplantation, the autograft site should be marked with nonabsorbable suture. The optimal site for autotransplantation should be left to the discretion of the individual surgeon taking patient-specific factors into account. (Strong recommendation, low-quality evidence.)

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Transcervical Thymectomy

Transcervical thymectomy can be performed during operations for kidney-related HPT due to the high prevalence of parathyroid tissue residing in the thymus. Intrathymic parathyroid glands are not uncommon: 5% to 30% of patients with kidney-related HPT have supernumerary parathyroid glands and 12% to 39.3% of patients have ectopic glands, and both are most commonly located in the thymus.[284,309–317] In addition, parathyroid cell rests are found in up to 37% of patients, and the most common location is also the thymus. The prevalence of intrathymic parathyroid tissues in the kidney-related HPT population ranges from ranges from 14.8% to 45.3%.[316,318–320] In the setting of kidney-related HPT, intrathymic parathyroid tissue will be affected by the same pathophysiologic process causing hyperplasia and therefore is a source for potential persistent or recurrent disease.[319] Consequently, some surgeons perform bilateral transcervical thymectomy as a standard therapy accompanying either subtotal resection or total resection with autotransplantation.[270,284,319–321] Other surgeons only perform transcervical thymectomy unilaterally if an ipsilateral gland cannot be located.[260,298,322] In general, thymectomy is not performed after total PTX without autotransplant, as patients rely on intrathymic parathyroid tissue for parathyroid function.

The superior pole of the thymic tongue is identified behind the sternothyroid muscle and inferior to the lower pole of the thyroid. The tissue is often a brighter yellow color. The superior pole is freed circumferentially, and the thymus is gently pulled anteriorly and superiorly in a hand-over-hand manner.[166] The thymic vessels are exposed in this manner and are carefully ligated with ties, clips or a vessel-sealing device, facilitating delivery of additional thymic tissue into the operative field. This can be performed on both sides of the neck, allowing for resection of one or both cervical thymic tongues.[319] Much of the data on outcomes following transcervical thymectomy is retrospective, or does not differentiate between patients with kidney-related hyperplasia and familial multigland disease. One study that evaluated a subset of patients with kidney-related parathyroid hyperplasia found an increased cure rate when bilateral transcervical thymectomy was performed compared to when it was not (94.0% vs. 89.0%).[321] Another retrospective study investigating patients undergoing reoperative neck surgery for recurrent disease reported intrathymic parathyroid glands in 28.4% of patients.[323] Based upon the desire to avoid reoperation, the European Society of Endocrinologists recommends transcervical thymectomy at the time of subtotal PTX or total PTX with autotransplantation.[267] Postoperatively, patients receiving transcervical thymectomy may have higher rates of transient hypocalcemia. There was no difference in the rate of other surgical complications between patients who did and did not receive transcervical thymectomy.[319]

Recommendation 6-6: Transcervical thymectomy should be performed in SHPT patients receiving subtotal PTX or total PTX with autotransplantation to reduce the likelihood of persistent or recurrent disease. (Weak recommendation, low-quality evidence.)

Comparison of 3 Surgical Approaches and Risks of Recurrent and Persistent Disease

Subtotal PTX Versus Total PTX With Autotransplantation. Subtotal PTX and total PTX with autotransplantation are the most commonly performed operations for SHPT. Despite the popularity of these 2 approaches, just 1 randomized control trial has been performed. This study by Rothmund et al[324] of 40 patients reported a higher rate of reoperation in patients receiving subtotal PTX than total PTX with autotransplantation (10.0% vs. 0.0%). In addition, patients receiving total PTX with autotransplantation had more significant symptomatic improvement, leading the authors to recommend total PTX with autotransplantation.[324] In contrast, other studies have demonstrated similar improvements in quality of life and symptom reduction following either operation.[325,326] Reviews of the American College of Surgeons National Quality Improvement Project data comparing patients who received a total PTX with autotransplantation to patients who received a subtotal PTX demonstrate that total PTX with autotransplantation is associated with a significantly longer operative time and prolonged hospital stay; other 30-day complication rates, including readmission rate and mortality, are similar.[257,327] Two meta-analyses have synthesized the published literature on these 2 operations. One meta-analysis of 5 studies reported similar recurrence rates between the 2 operations, and concluded that the choice of operation should be left to the surgeon.[328] More recently, a meta-analysis of 18 studies compared 1864 patients receiving a subtotal PTX to 1792 patients receiving a total resection with autotransplantation, and determined the 2 operations had similar rates of symptomatic improvement (93.0% for subtotal PTX vs 89.0% for total PTX with autotransplant), radiologic changes (85.4% and 85.3%), postoperative hypocalcemia (16.6% and 18.1%), persistent disease (6.1% and 2.0%), recurrent disease (9.2% and 7.1%), and reoperation (5.3% and 5.8%), as well as time to recurrence.[329] However, patients who underwent total PTX with autotransplantation had a longer operative time, longer postoperative length of stay, a lower calcium level at 1 month postoperatively, and a higher requirement for vitamin D supplementation 1 year postoperatively.[329] This author group reached the same conclusion: as SHPT was successfully treated in both groups with similar recurrence rates, the choice of operation should be left to the surgeon.[329]

Total PTX With Autotransplantation Versus Total PTX. One multicenter randomized control trial has compared total PTX with autotransplantation and transcervical thymectomy to total PTX without autotransplantation and transcervical thymectomy and followed the patients for 3 years postoperatively. There were no differences in operative duration, postoperative length of study, calcium or PTH levels at discharge, the number of patients needing calcium and vitamin D at discharge, or long-term mortality between the 2 groups.[290] Persistent and recurrent disease occurred at lower rates following total PTX than total PTX with thymectomy and autotransplantation (1.9% and 0.0% vs 4.1% and 8.3%).[290] A greater proportion of total PTX patients required calcium and vitamin D supplementation than those with autotransplantation and thymectomy at the 3-year follow-up point, but none required autotransplantation of cryopreserved tissue. Consequently, the authors concluded that autotransplantation was not necessary.[290] Three systematic reviews or meta-analyses have analyzed the collective published data on these operations. Two studies concluded that there is no difference in the rate of postoperative hypoparathyroidism,[274] while 1 study reported lower rates of hypoparathyroidism following autotransplantation.[330] Otherwise, each has concluded that there is no difference in the rates of postoperative complications or symptomatic improvement between the 2 operations.[274,330] One study reported no difference in the rate of persistent disease following either operation.[284] All 3 concluded

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that total PTX with autotransplantation is associated with higher rates of recurrent disease and need for reoperation than total PTX alone.[274,330]

Comparison of 3 Approaches. The 3 most common operative approaches have rarely been compared collectively. Retrospective studies report that all 3 operations result in similar symptomatic improvement and similar decreases in both calcium and PTH level in the immediate postoperative period.[266,280] The only randomized controlled trial for the 3 operations enrolled 63 patients and concluded that subtotal resection was associated with the shortest operation and shortest length of stay as well as the lowest rates of hypocalcemia, but the highest rates of recurrent disease.[331] Two small single-institution retrospective studies reported increased rates of disease recurrence following subtotal and total PTX with autotransplantation compared with total PTX without autotransplant (recurrence rates 18.0%–20.8% vs 0%–4.5%).[278,280] However, these 2 studies evaluated 48 and 94 patients each.[278,280] The largest retrospective study evaluated 606 patients, although the vast majority (504) received a total PTX with autotransplantation, compared with 21 patients receiving a subtotal PTX and 32 patients receiving a total PTX (49 patients had an "incomplete" resection, defined as fewer than 4 parathyroid glands identified). The rates of persistent and recurrent disease were the lowest after total PTX alone (0.0% and 0.0%), followed by total PTX with autotransplantation (0.4% and 4.5%); higher rates of persistent and recurrent disease were noted after subtotal resection (4.8% and 9.5%).[266] It is notable that these differences did not reach statistical significance, and the authors noted that reduced experience with subtotal resection may have inadvertently led to higher recurrence rates.[266] In addition, this study did not assess long-term hypoparathyroidism, transient or long-term hypocalcemia, postoperative calcium or vitamin D supplementation needs.

Total PTX without autotransplantation may not be the optimal surgical approach for many patients. A network meta-analysis evaluated the rates of persistent and recurrent disease for all three operative strategies. Twenty-six studies with 5063 patients were included, many of which are cited above. Subtotal PTX was most likely to result in recurrent disease and reoperation, followed by total PTX with autotransplantation and then total PTX alone.[332] However, total PTX was the most likely to result in hypoparathyroidism, leading the authors to recommend total PTX with autotransplantation as the most "efficient and safe" operation.[332]

Recommendation 6-7: Subtotal PTX is the preferred operation for patients with SHPT and THPT when all parathyroid glands are identified and enlarged. However, the risks of persistent and recurrent HPT and postoperative hypocalcemia should be weighed; the choice of operation should be left to the individual surgeon and be based on each individual patient's circumstances. (Weak recommendation, moderate-quality evidence.)

THPT: Operative Principles

The intraoperative goal of surgery for THPT is to decrease the amount of hyperfunctioning parathyroid tissue.[333] As with surgery for SHPT, there is debate about the optimal surgical procedure to accomplish this goal.

Limited Resection

The presence of single or double adenomas rather than 4-gland hyperplasia has been reported in 2.6% to 32% of patients with THPT.[264,271,334–337] While the pathologic basis behind morphologically and functionally different parathyroid glands within 1 patient is unknown, variation in the size of glands within a patient is widely acknowledged.[333,336] Consequently, some author groups have advocated for "selective" or "limited" resections of the 1 or 2 abnormal glands.[335] A selective or limited resection is performed as follows: a bilateral neck exploration is performed, and all 4 glands are identified. If only 1 or 2 abnormally enlarged glands are identified and the other glands are normal in appearance, then only the abnormal glands are resected.[333,335,338] Only one study defines "abnormally large" as >50 mg.[338] Because the majority of patients with THPT have multiple abnormal glands, a bilateral exploration (not a unilateral exploration) should be performed, regardless of preoperative imaging results.[336] If enlargement of 3 or 4 glands is noted, a subtotal or total PTX is indicated; if fewer than 4 gland are identified, limited resection is only recommended if other glands are normal in size.[338]

Recommendation 6-8: In patients undergoing initial PTX for THPT, bilateral exploration should be performed with identification of all four parathyroid glands. Limited resection can be indicated if only 1 or 2 glands is abnormally enlarged. (Weak recommendation, low-quality evidence.)

Subtotal PTX

Subtotal PTX consists of a bilateral neck exploration and resection of all parathyroid tissue except for a vascularized remnant gland which is then marked with a permanent suture, clip, or both. The most "normal" appearing (without nodularity) or smallest gland should be chosen for subtotal resection.[322,339] Various author groups have proposed leaving behind a remnant the size of 1 to 2, 2, or 2 to 3 times a normal gland size, or a remnant weighing ~30 to 50, 40 to 50, 40 to 80, or 50 to 75.[131,264,336,338–342]

Recommendation 6-9: In THPT patients undergoing surgery, subtotal PTX should leave behind a well-vascularized remnant of the most normal-appearing gland. (Strong recommendation, low-quality evidence.)

Total PTX With or Without Autotransplantation

Total PTX for tertiary disease is carried out the same as in patients with secondary disease: bilateral neck exploration and resection of all parathyroid tissue. Most institutions perform a simultaneous autotransplantation.[284,334] Two studies have reported outcomes following total PTX without autotransplantation or thymectomy. For all patients studied, the postoperative PTH levels remained below normal for several years, and these patients required long-term vitamin D or calcium supplementation.[343,344]

Recommendation 6-10: Total PTX without autotransplantation should not be performed in patients with tertiary HPT. (Strong recommendation, low-quality evidence.)

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Transcervical Thymectomy

Transcervical thymectomy has not been specifically studied in this patient population. One study reported that 29% of patients with THPT had an intrathymic gland, and therefore recommended thymectomy as a means to lower the rate of recurrent disease.[339] Other authors recommend performing thymectomy only if an inferior gland is not found in the expected location, if an intrathymic gland is suggested based upon preoperative imaging or radioguided exploration, or if the patient is at higher risk of recurrent disease (such as a young patient with a longer life expectancy).[128,338]

Recommendation 6-11: Transcervical thymectomy may be considered in select patients with THPT. (Weak recommendation, low-quality evidence.)

Comparison of Surgical Approaches

There have been no randomized controlled trials comparing the 3 surgical approaches for THPT (limited resection, subtotal resection, and total PTX with or without autotransplant) with respect to persistent or recurrent disease or postoperative hypocalcemia. Furthermore, the definitions of persistent or recurrent disease are inconsistent in various studies: some use calcium levels only, while others factor in PTH levels as well. A recent systematic review explored the published data on rates of persistent and recurrent disease following PTX for THPT. The study included 240 patients with information about persistent and recurrent disease, representing the largest THPT patient cohort studied. Persistent HPT was reported in 1.3% and recurrent HPT in 7.6% of patients undergoing subtotal PTX.[116] Of patients receiving total PTX, 0.0% of patients experienced persistent disease and 4.0% experienced recurrent disease.[116] Retrospective single-institution studies have compared the outcomes following selective or limited resection to subtotal or total PTX. Some groups identified similar low rates of persistent or recurrent disease following limited resections (0.0%–6.7%) and subtotal or total parathyroidectomies (0.0%–8.1%); because of low numbers of patients included in these studies, there were no significant differences between any cohorts.[123,333,335,338] These studies all used hypercalcemia to identify persistent or recurrent disease. In contrast, a 2-institution study determined that limited PTX increases the risk of persistent disease; this group utilized PTH levels adjusted for kidney function to determine success.[345] Patients who underwent limited resections as compared to total or subtotal resections had similar operative times, lengths of stay, and resected parathyroid gland weights.[338] Lower rates of transient and persistent hypocalcemia following limited resection (5.0% and 0.0%) than following subtotal/total PTX (27.0% and 7.0%) have been reported.[335,338]

Recommendation 6-12: In patients with THPT, the choice of operation should be left to the individual surgeon, who should incorporate the risks and benefits of each operation into decision-making. (Strong recommendation, low-quality evidence.)

Role of ioPTH Monitoring

While ioPTH monitoring has become standard in PTX for PHPT, the role of ioPTH monitoring during PTX for SHPT and THPT is less clear. The reasons for this uncertainty are 2-fold: (1) surgeons inherently must perform a bilateral neck exploration for most patients with kidney-related HPT, and (2) there is impaired metabolic clearance of PTH in patients with kidney failure, potentially confounding the ability to interpret and apply ioPTH in real time. This uncertainty is reflected in practice patterns: in a 2013 international survey with 86 respondents, 36 (46.2%) surgeons routinely used ioPTH , while 19 (24.4%) utilized ioPTH monitoring only during reoperations or for scientific purposes.[281]

Secondary HPT: Criteria for ioPTH Decline and Utility of ioPTH Monitoring

The half-life of PTH in patients with PHPT is well-established, leading to recommended criterion for expected PTH decay following resection of adenomatous or hyperplastic glands.[1] Because of the impaired renal and hepatic clearance of (1–84) PTH and its fragments in patients with kidney failure, however, the PTH decay is less predictable in this population. While some literature has demonstrated that there is no difference in the relative decrease of ioPTH in CKD and non-CKD patients, several other studies have demonstrated a slower decay and longer half-life of ioPTH in patients on hemodialysis compared to those with normal kidney function.[300,346–349] Multiple studies have evaluated criteria for decay of ioPTH to predict a successful resection. However, different surgical techniques and varying definitions of postoperative cure have been utilized, limiting the ability to compare results from different studies and gain cohesive understanding. Studies reporting various protocols for the drawing of ioPTH levels and goal criteria for ioPTH decreases are summarized below in Table 5.

The utility of ioPTH monitoring in the setting of CKD and ESKD is still uncertain, as a bilateral exploration is inherently required. One group concluded that ioPTH monitoring was of no value during operations for secondary disease, as ioPTH levels did not impact surgical decision-making.[356] In addition, while ioPTH decreases were accurate in predicting cure, the ioPTH decay was unable to determine persistent HPT.[355,360,361] In addition, ioPTH testing was incapable of determining if all parathyroid tissue was resected and therefore if autotransplantation was needed.[346,355,361,362] Other authors concluded that ioPTH monitoring was useful.[354] Two large studies determined that ioPTH monitoring changed intraoperative plans in 13.1% to 15.7% of patients, either by indicating the presence of residual parathyroid tissue following resection of 4 glands, or by indicating that fewer than 4 glands were present.[352,363] The rates of recurrent and persistent disease are lower when ioPTH monitoring is used when compared with when it is not.[340] A recent survey indicated that 38.2% of surgeons employing ioPTH used it to guide further surgical exploration.[281] Two studies have evaluated the benefit of postoperative PTH levels to predict success or failure; each has investigated patients following total PTX with autotransplantation.[352,353] However, as these levels are drawn after the conclusion of the operation, their utility to guide intraoperative decision-making is poor. Multiple studies have demonstrated that postoperative hypocalcemia is not predictive of operative success.[353,364]

Recommendation 6-13: The benefits and limitations of ioPTH monitoring in patients with SHPT should be understood by all surgeons performing these operations. The limitations may outweigh the benefits in most situations. (Weak recommendation, moderate-quality evidence.)

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Tertiary HPT: Criteria for ioPTH Decline, and Utility of ioPTH Monitoring

Patients who have undergone kidney transplantation generally have improved kidney function, and therefore theoretically a more predictable decline in ioPTH postresection. However, several studies have reported slightly delayed clearance of ioPTH in patients with THPT.[300,341,359] This is likely due to a decrease in GFR over time following kidney transplant that results in the increased production and accumulation of PTH.[342] Consequently, the literature surrounding the utility of ioPTH assays in patients with THPT is somewhat more robust than that for patients with secondary disease, but is not definitive. In addition, there have been relatively few publications that focus solely on patients with tertiary disease. Application of the Miami criterion, drop of ioPTH by half from the baseline level, has been determined in several studies to be highly sensitive to predict cure for THPT, which was defined as normocalcemia post-PTX.[128,342,356,359,365,366] Not all patients may experience a decrease in 50% within 10 minutes, so expansion of the timeframe to 15 or 25 minutes may include more patients while maintaining >89% sensitivity to predict cure.[341,356] An alternative approach is a PTH target, which is also often employed during PTX. A final ioPTH of <200 pg/mL is the most sensitive for predicting long-term normocalcemia; more stringent criteria of a final ioPTH of <60 pg/mL or a 50% decrease and into the normal range had lower sensitivities of 68% and 66%, respectively.[341]

While all of the above studies have demonstrated a high sensitivity and/or positive predictive value for cure, none report a high specificity or negative predictive value. There is no ioPTH percentage drop noted that successfully predicts failure. Because of the lack of well-established PTH targets, low overall failure rates, and patients attaining cure without meeting ioPTH decrease criteria, some authors ultimately conclude the utilization of ioPTH during PTX for tertiary disease is not critical.[341,342] Conversely, other author groups concluded that ioPTH testing impacted the real-time operative care of THPT as ioPTH assays allow for limited rather than subtotal resections in 9% to 20% of patients.[356,359] Failure to decline may indicate the presence of ectopic or supernumerary glands.[358,359] Lastly, the rates of persistent and recurrent disease declined when ioPTH was employed during PTX when compared with operations where ioPTH was not used.[340]

Recommendation 6-14: The benefits and limitations of ioPTH monitoring in patients with THPT should be understood by all surgeons performing these operations. The benefits may outweigh the limitations in most situations. (Weak recommendation, moderate-quality evidence.)

Type of Assay Used

PTH is an 84 amino acid polypeptide chain known as (1–84) PTH. (1–84) PTH and its fragments are degraded by the kidneys and the liver. In patients with CKD, both renal and hepatic clearance are impaired.[367,368] Some PTH fragments, in particular the C-terminal 7 to 84 fragment, are cleared primarily through glomerular filtration, and thus accumulate in patients with CKD and ESKD and have a longer half-life.[352,369] The ability of an ioPTH assay to accurately detect (1–84) PTH and not its fragments affects the accuracy of ioPTH monitoring. The first generation of ioPTH assays measured only one fragment of the PTH polypeptide chain, and thus were inaccurate due to high cross-reactivity.[367,370] Assays that measured 2 segments of the PTH molecule—the N-terminus and the C-terminus—were therefore developed, and are known as "intact" assays. The Roche intact immunoassay utilizes 2 monoclonal antibodies: 1 to amino acids (AAs) 26 to 32 and the other to 55 to 64AA, and has been shown to have a strong correlation to the standard immunofluorimetric assay.[371] The Nichols intact immunoassay uses 2 polyclonal antibodies to 12 to 34AA and 39 to 84AA. However, each of these has a high degree of cross-reactivity with (7–84) PTH fragments: the Roche assay has a 68% cross-reactivity to (7–84) PTH, while the Nichols immunoassay has a 100% cross-reactivity.[354,367] The (7–84) PTH fragments that accumulate in CKD and ESKD may contribute to and falsely elevate the ioPTH result, confounding interpretation.[348] A third-generation "whole" ioPTH assay has no cross-reactivity to (7–84) PTH fragments. The published data on the accuracy of the whole ioPTH assay is conflicting.[349,351,367] Furthermore, the whole ioPTH assay is no longer commercially available.[352] The ioPTH assays currently employed are generally the second generation "intact" ioPTH tests.

Recommendation 6-15: When using ioPTH, a second-generation ioPTH assay should be employed. (Strong recommendation, high-quality evidence.)

Definition and Risks of Adynamic Bone Disease

Low bone turnover disease, also known as adynamic bone disease, is associated with increased risk of fractures and cardiovascular morbidity and mortality. Adynamic bone disease is diagnosed with bone biopsy, but due to the invasiveness of the procedure many clinicians utilize PTH and bone-specific alkaline phosphatase as markers of the condition. Long-standing hypoparathyroidism is one potential contributor to adynamic bone disease, although other factors including age and diabetes mellitus also play a role.[372] The association between PTX and adynamic bone disease has only been studied in small patient cohorts.[373–376] The "complete cessation" of bone mineralization following PTX and development of adynamic bone disease in 90% of PTX patients on dialysis within 1 year postoperatively have both been reported.[375,376] In contrast, a prospective matched cohort study of hemodialysis patients indicates that although the risk of fractures transiently increases in the 90 days following PTX, the overall risk of hip and femur fractures is decreased compared with hemodialysis patients not undergoing PTX.[86] Some authors suggested a target postoperative PTH in the range recommended by the National Kidney Foundation KDOQI and KDIGO.[377] No studies have been carried out that investigate the association between the postoperative PTH level and the risk of fracture, adynamic bone disease, or cardiovascular disease following PTX.

Cryopreservation

Cryopreservation allows for metachronous autotransplantation of parathyroid glands, which are harvested at the time of surgery, in the event of long-lasting hypoparathyroidism. The process of cryopreservation and thawing have been described elsewhere, and are beyond the scope of this text.[259] Similar to immediate autografts, the most common sites of reimplantation are the forearm and the sternocleidomastoid muscle.[281] The majority of the published literature on cryopreservation discusses the role of cryopreservation in patients with PHPT. There are no commonly accepted indications for the use of cryopreservation in

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TABLE 5. Studies Evaluating ioPTH Use in Secondary and Tertiary Hyperparathyroidism

References	No. Patients	Type of HPT	Type of Surgery	Cure Goal/Conclusion
Chou et al[350]	24	SHPT	Total + ATX	60% decrease at 10 min
Freriks et al[351]	42	SPHT	Total	> 70% at 10 min
Hiramitsu et al[352]	226	SPHT	T + ATX + TCT	> 70% at 10 min; ioPTH <60 on POD #1
Nascimento et al[353]	53	SHPT	T + ATX	> 95% decrease from baseline within 1st week postop, or <ULN within 1st week postop
Seehofer et al[354]	153	SHPT	Total + ATX, subtotal	PTH <150 pg/mL at 15 min
Vulpio et al[355]	42	SHPT	Total + ATX, subtotal	PTH <166 pg/mL and <12% baseline at 20 or 30 min
Lokey et al[347]		SHPT, THPT	Subtotal	> 50% at 20 min from preoperative baseline
Ohe et al[300]	86	SHPT, THPT	Total + ATX	80% decrease at 20min
Pitt et al[356]	33	SHPT	Total + ATX, subtotal	> 50% by 25 min is predictive of postop cure
	72	THPT	Subtotal	> 50% by 25 min
Weber (2005)[357]	95	SHPT, THPT	Total	Normal level at 15 min is predictive of cure
Casarim et al[358]	33	THPT	T + ATX	> 71.2% at 20 min
Haustein et al[359]	32	THPT	Subtotal	> 50% at 10 min
Triponez et al[342]	35	THPT	Subtotal + TCT	Decrease > 50% at 10 min

ATX indicates autotransplantation; POD, postoperative day; TCT, transcervical thymectomy; ULN, upper limit of normal.

patients with SHPT or THPT. Cryopreservation is performed for a variety of reasons: (1) during a total PTX without autotransplantation due to the potential for hypoparathyroidism; (2) regardless of operation type, in case of autograft or remnant gland failure; and (3) in case of a future reoperation that may render a patient hypoparathyroid.[260,266,270,281,284] Other surgeons, however, report no cases of permanent hypoparathyroidism following total PTX with or without autotransplantation, and consequently recommend against cryopreservation.[290]

Surgeons who perform cryopreservation report infrequent implantation in patients with SHPT or THPT.[281,378] One contributing factor to this low rate of reimplantation is the lack of guidelines on the duration or severity of postoperative hypoparathyroidism that indicates the need for autotransplantation of cryopreserved glands. One author suggests implantation if "profound" hypoparathyroidism exists at least 6 months postoperatively, but provides no data to support this recommendation; furthermore, assessment of hypoparathyroidism may be clouded by ongoing HBS.[166] Other factors suggested include the severity of symptoms, the need for calcium and vitamin D supplementation, and the presence of adynamic bone disease seen on histology.[378] In addition, there are conflicting data about the viability of cryopreserved parathyroid glands. Of implanted cryopreserved parathyroid glands, only 10% to 46% become fully functional, while 10% to 23% are partially functional, meaning some parathyroid function exists but calcium supplementation is still required.[259,379] While one center has reported a 100% viability rate, others report 31% to 100% of cryopreserved glands are nonfunctional.[259,378,379] The functionality of implanted cryopreserved glands may also be related to the number of reimplanted fragments.[378] Furthermore, shorter cryopreservation time is associated with improved gland functionality.[259,379] Importantly, recurrence of HPT has been reported in cryopreserved autografts.[281]

Recommendation 6-16: Cryopreservation may be considered in patients receiving a total PTX. (Weak recommendation, low-quality evidence.)

OTHER INTRAOPERATIVE ADJUNCTS AND NONSURGICAL APPROACHES

Intraoperative Ultrasound

Intraoperative ultrasound is a readily accessible, noninvasive, and cost-effective adjunct for localization of parathyroid glands. Since SHPT tends to be a longstanding disease process before it becomes medically refractory requiring PTX, the involved parathyroid glands are larger and more easily identified on ultrasound. There is some evidence that the maximum diameter of the parathyroid gland (> 8 mm) is associated with the presence of nodular hyperplasia.[26,380] Intraoperatively, ultrasound can be used to discern gland location with the patient in the hyperextended, intubated position. Use of smaller footprint probes can facilitate imaging of unique locations that would otherwise be unavailable.

Frozen Section Analysis

As previously described, frozen section analysis can be used intraoperatively to identify or exclude the presence of parathyroid tissue.[1] However, similar to the case of PHPT, results can be misleading. Although in some cases, scarce intraparathyroidal fat can suggest hypercellularity, in general, histologic cellularity does not reliably discern either adenoma from multigland disease or normal from abnormal parathyroid glands, and therefore frozen section analysis is not always a reliable intraoperative parameter.

Parathyroid Aspiration

Although frozen section is the most commonly used intraoperative adjunct to confirm the identity of parathyroid tissue, it can be a time-consuming process, and occasionally confirmation cannot be made without the use of permanent section such as in the case of intrathyroidal lesions. When ioPTH is already being used, direct aspiration of the parathyroid gland allows for an expeditious and cost-effective confirmation of parathyroid tissue by using the same ioPTH immunoassay. This technique accurately confirms parathyroid tissue with a sensitivity of 97% to 100%, specificity of 100%, and has 100% concordance with frozen section analysis.[381–383]

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Probe Radioguidance

The use of injecting Technetium 99-m as a radioisotope to help localize and confirm hyperfunctioning parathyroid tissue has become a popularly used intraoperative adjunct in PTX for PHPT as described in the AAES guidelines for PHPT.[1] Use of the Technetium 99-m and the radioguidance probe can facilitate identification of hyperfunctioning parathyroid glands including those in ectopic locations and has been shown to decrease overall operative time.[384–386] Use of the " > 20% ex vivo count" rule also identifies both hyperplastic and adenomatous parathyroid glands with high sensitivity, and has been suggested as a reliable substitute for frozen section analysis.[337,366,387]

Fluorescence Imaging

Fluorescence imaging is an instant on-table method for direct visualization of parathyroid tissue known to exhibit increased autofluorescence intensity when exposed to near-infrared light, up to 8.5-fold higher than surrounding tissue including the thyroid gland. Near-infrared light can detect parathyroid tissue with a sensitivity of 83% to 92% and is gaining popularity as a useful surgical adjunct for intraoperative identification of parathyroid tissue.[388–391]

RLN Monitoring

The use of RLN monitoring in the setting of PTX was comprehensively discussed in the AAES guidelines for PHPT.[1] In summary, there is a dearth of data on the use of RLN monitoring in the setting of PTX, and although there may be some utility in re-operative cases, there is no evidence that RLN decreases the rate of injury in initial PTX for PHPT. There are even less data evaluating the utility of RLN monitoring in PTX for SHPT or THPT.

Energy-based Vessel Sealing Devices

Total thyroidectomy and PTX were originally operations performed with the classic "clamp and tie" technique for dividing blood vessels. However, in the last couple of decades, advanced energy-based vessel sealing devices have been introduced to facilitate control of vascular pedicles and have become widely used. The most commonly used devices for thyroidectomy and PTX are the Harmonic Focus (HF; Johnson and Johnson, Ethicon *Endo-surgery*, Cincinnati, OH) and the Ligasure Small Jaw (LSJ;*Covidien, Boulder, CO*). Use of these devices have been shown to decrease operative time without any significant difference in surgical complications including rates of hematoma or nerve injury.

Remote Access Approach

Although a bilateral neck exploration is the most common operative technique used for PTX, other remote access techniques (approaching the thyroid and parathyroid glands from more distant sites to avoid a cervical neck scar) have been used in select patients with sporadic PHPT and localized adenomas.[1] These remote access approaches are less suitable for PTX in the setting of SHPT or THPT for which the standard operative technique is a bilateral neck exploration.

Percutaneous Ethanol Injection or Ethanol Ablation (EA)

Percutaneous ethanol ablation of parathyroid glands suspected to have nodular hyperplasia was first introduced in Europe in the 1980s.[17,18] By the 2000s, selective EA was widely adopted in Japan for treatment of advanced SHPT that was refractory to medical therapy. EA is ideal for glands with suspected nodular hyperplasia that are identified on ultrasound evaluation. A 22-G needle is advanced into the middle of the parathyroid gland under image guidance, and an appropriate amount of ethanol is slowly injected to create a hyperechoic bubble covering ~80% of the parathyroid gland volume (0.02–0.1 mL). Serum PTH levels should be monitored 2 to 4 weeks after the injection to determine if target PTH levels were reached. Additional EAs can be considered if target serum PTH levels are not met, with special attention to target any areas of the gland that have increased vascularity. However, the number of EA sessions should be limited overall to prevent undue sclerosis and adhesions that may make subsequent operations difficult. If serum PTH levels remain elevated despite an ultrasound that demonstrates complete destruction of the parathyroid gland, imaging to evaluate for ectopic gland is indicated. It is important to know that EA can cause pain, hematoma, and leakage of ethanol can cause RLN injury, as well as scarring of adjacent tissue. Studies evaluating the efficacy of EA as an alternative treatment for patients with SHPT that is refractory to medical therapy show that it is able to maintain PTH at target levels after 1 year of treatment for 81.2% of patients with only 1 or 2 parathyroid glands with nodular hyperplasia.[392–394] The efficacy of EA decreases with an increase in the number of enlarged parathyroid glands.[392,395]

Thermal Ablation

Percutaneous thermal ablation techniques such as radiofrequency, microwave, and laser ablation have also been used as alternative treatments to PTX for patients with SHPT.[396–405] Similar to EA, only enlarged parathyroid glands that are fully visualized on ultrasound are candidates for these techniques. All of these techniques utilize electrical currents that use heat to cause tissue necrosis in target tumors. More widely adopted in Asia, these thermal ablation techniques have been shown to be a potentially good alternative treatment to PTX. A recent meta-analysis of four randomized control trials and 2 cohort studies involving a total of 326 patients demonstrated no significant difference in serum PTH, calcium, or phosphorus between patients treated with thermal ablation or those treated with PTX. However, thermal ablation was associated with a decreased risk for hypocalcemia (OR: 0.23, 95% CI: 0.11–0.47, $P < 0.01$) and a higher risk of recurrence (OR: 4.24, 95% CI: 1.44–15.76, $P = 0.03$).[406] The most commonly reported complications include hematoma and transient nerve injury. Noninvasive high intensity focused ultrasound uses a 60 mm ultrasound probe to emit sonographic waves that causes cavitation and eventual necrosis in target tumors. Kovatcheva et al[407] reported a pilot study applying this technique to 5 patients with SHPT due to CKD. They were able to demonstrate transient decreases in serum PTH, but ultimately were unable to sustain target PTH levels in the long term. Complications included local edema as well as transient impairment of vocal cord mobility.

Cryoablation

Instead of heat, cryoablation uses liquid nitrogen or argon gas to create intense cold to freeze and destroy tissue within the target tumor. There are case reports that describe the use of image-guided cryoablation of parathyroid glands, but it has not been a widely adopted technique.[406,407]

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OUTCOMES OF PTX FOR SHPT AND THPT

Overall Postoperative Complication Rate

The overall major complication rate following PTX for SHPT and THPT in the modern era ranges from 3.8% to 15.1%.[257,266,327,356] The 30-day readmission rate is 17%, which is far higher than the <1% rate of readmission following PTX for PHPT.[1,327,410] The high rates of postoperative complications and readmission indicate the need for coordinated multidisciplinary care in the preoperative and postoperative settings.[257] Patients should be appropriately counseled about the risks of major postoperative complications.

Postoperative Hypocalcemia and Treatment Protocols

Postoperative hypocalcemia is extremely common in patients with SHPT and THPT due to both the metabolic derangements stemming from ESKD and HBS. HBS describes a state of unopposed osteoblast uptake of calcium after a decrease in PTH, and consequently patients with HBS experience severe and prolonged hypocalcemia.[198] HBS is defined as a decrease in serum calcium to <8.4 mg/dL or hypocalcemia for more than 4 days postoperatively.[198] Patients may experience classic signs and symptoms of perioral and digital paresthesia and muscle cramps; more severe signs include tetany, seizures, myocardial dysfunction, and electrocardiographic conduction abnormalities. Calcium levels reach their nadir ~48 to 72 hours postoperatively, and patients must be treated appropriately to avoid development of life-threatening hypocalcemia. Transient hypocalcemia is reported in 15% to 59% of patients undergoing PTX for secondary disease.[260,261,264,298] Hypocalcemia can result in outpatient visits, emergency room visits, and readmissions to the hospital.[411,412] Risk factors for hypocalcemia include younger age, higher weight, higher preoperative alkaline phosphatase, and lower preoperative calcium levels.[264,413–417] Data regarding an association between higher preoperative PTH levels and postoperative HBS are conflicting.[418] Based upon a large meta-analysis, there is no difference in the rate of postoperative hypocalcemia based upon operation type, although patients receiving total PTX with autotransplantation may require calcium and vitamin D supplementation for longer durations than patients receiving subtotal PTX.[329]

Postoperative hypocalcemia is less severe in patients with tertiary disease than in patients with secondary disease.[419] Transient postoperative hypocalcemia develops in 0% to 59% of postkidney transplant patients who have subsequently undergone PTX.[333,335,342,366] While hypocalcemia can be rare following subtotal PTX for THPT, it may occur in 76% to 100% of patients following total PTX with autotransplantation.[128,307] Permanent hypocalcemia occurs in 0% to 27%, and patients may require implantation of cryopreserved tissue.[122,333] There are no described treatment protocols for hypocalcemia following PTX for SHPT or THPT.

Because of the potential severity of postoperative hypocalcemia following PTX for SHPT, patients are monitored and treated aggressively. A number of treatment protocols for secondary disease have been reported in the literature, but there are no comparisons of their efficacy. The majority of published treatment protocols initiate oral calcium and calcitriol administration immediately postoperatively, if not already started in the preoperative setting, and monitor serum calcium levels regularly (Table 6).[69,166,260,266,277,420,421] Intravenous calcium gluconate is generally added when serum calcium levels fall below a specific threshold, although some advocate for immediate administration based upon alkaline phosphatase levels.[198,266,277,420,421] For patients on hemodialysis, a high calcium bath can also be employed.[372,419] Calcimimetic medications should be immediately discontinued due to the risk of hypocalcemia.

Because of bone formation occurring in the setting of HBS, patients may develop hypomagnesemia, hypophosphatemia, and hyperkalemia, in addition to hypocalcemia.[198] Magnesium should be supplemented as needed. Phosphate binders should be discontinued if the serum phosphorus levels decrease, and phosphorus supplementation initiated if indicated.[198]

Recommendation 8-1: Postoperative hypocalcemia should be treated with oral and intravenous calcium supplementation, as needed. Postoperative calcitriol is also indicated. Calcimimetics should be discontinued. (Strong recommendation, low-quality evidence.)

Postoperative Length of Stay

While PTX for primary disease is often performed as an outpatient operation, PTX for kidney-related HPT generally results in admission for several days postoperatively due to the need to stabilize calcium levels. Approximately 1-quarter of these patients are admitted to an intensive care unit postoperatively.[411] Encouragingly, the mean length of stay postoperatively is decreasing over time.[422] The mean postoperative length of stay following PTX for SHPT ranges from 4 to 10 days.[257,280,290,298,411,412,419,420] Some studies have demonstrated a shorter length of stay following subtotal resection compared with total PTX with autotransplantation, but others found no difference.[257,258,280,410] In contrast, many centers perform PTX for THPT as outpatient procedures. The mean length of stay following an operation in a kidney transplant recipient is significantly lower, at 2.2 to 3.2 days.[338,339,419] Limited data demonstrates no difference in the postoperative length of stay following subtotal versus limited resections.[338] Patients should be counseled preoperatively about the anticipated duration of hospitalization postoperatively depending on their disease.

Readmission

Following PTX for SHPT, 4-17% of patients are readmitted within 30 days.[257,410] There is no consistent evidence indicating that readmission rates are higher following total PTX with autotransplantation than following subtotal PTX.[257,410] Hypocalcemia and electrolyte disturbances are the most common causes of readmission.[410,423]

Permanent Hypoparathyroidism

Permanent hypoparathyroidism occurs when all or nearly all parathyroid tissue is resected, rendering the patient without sufficient functioning parathyroid tissue. Consequently, patients may be permanently hypocalcemic, requiring regular calcium supplementation and/or calcitriol administration. Permanent hypoparathyroidism is reported following surgery in up to 9.2% of patients with kidney-related HPT.[260,264,289,320,424,425] Higher rates are reported following total PTX with autotransplantation than after subtotal resection, although this data is not robust.[271,289,424] In patients with postkidney transplant disease, permanent hypoparathyroidism or hypocalcemia is reported in up to 10.0% of patients.[116,122,215,265,322,336,338,342,356] Permanent hypoparathyroidism seems to occur more after total PTX with autotransplantation than after subtotal or limited resection.[116,128,322,338] Patients with permanent hypoparathyroidism should be considered for reimplantation of cryopreserved tissue if possible.

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Recommendation 8-2: Surgeons should counsel patients about the risk of permanent hypoparathyroidism. (Strong recommendation, low-quality evidence.)

RLN Injury

The rate of RLN injury is overall low. The reported incidence of temporary recurrent laryngeal palsy following PTX for SHPT or THPT ranges from 0.0 to 15.5%.[122,261,264,265,277,280,287,290,339,420,426] The rate of permanent RLN injury or injury requiring vocal cord injection is even lower, with rates up to 2.1%.[116,260,266,280,290,298,320,424] The rate of RLN injury during reoperation is not well reported; one study reporting 30 reoperations in the neck reported no RLN injuries.[427] Smaller studies also report no or rare postoperative hoarseness or nerve injury.[266,320,338,428] Surgeons should counsel patients preoperatively about the risk of RLN injury.

Bleeding

Postoperative bleeding resulting in hematoma can be a life-threatening complication of surgery in the central neck due to the potential for airway compromise. Postoperative hematoma requiring reoperation occurs in 0.0% to 2.1% of patients following PTX for SHPT and THPT.[264,266,272,277,290,327,412,420] Other single-institution studies report that postoperative bleeding requiring reoperation occurs at higher rates (2.8%–5.7%) in patients on therapeutic anticoagulation.[339,342] Rates of postoperative hematoma not requiring intervention are less commonly reported, but this occurs in 1.0% to 3.0% of cases.[280,297,424] There is a paucity of data comparing rates of postoperative hematoma or the need for urgent reoperation between the different surgical options.

Because of the potential life-threatening nature of postoperative cervical hematoma, patients should be monitored postoperatively. Signs and symptoms of postoperative hematoma include neck swelling, worsening neck pain, dyspnea, voice changes, and an impending sense of doom. Neck hematomas should be managed as per the AAES guidelines for PHPT: evacuation in a controlled operative setting with anesthesia if the patient is stable and in no distress, or emergent evacuation at the bedside followed by transfer to the operating room for definitive management if the airway is compromised.[1]

Recommendation 8-3: Patients should be monitored postoperatively due to the risk of postoperative cervical hematoma. The time of monitoring depends on the clinical situation. (Strong recommendation, low-quality evidence.)

Infection

Postoperative surgical site infection is uncommon following PTX. Rates of postoperative neck infection are ~1.1% to 1.3%.[266,424] The addition of surgical sites for transplant autograft placement inherently increases the potential for surgical site infections, and single-institution studies have reported wound infections occurring in 0.9% of patients with an autograft site in the arm, and in 2.1% of patients with an abdominal autograft site.[266] Postoperative pneumonia has also been reported in this patient population, occurring in 0.8% to 4.9% of patients in single-institution studies.[266,277,356,426] Surgeons should be aware of the potential for surgical site infection despite a clean case.

Cardiac Morbidity

Cardiovascular events occur at a higher rate in patients with SHPT due to underlying ESKD. In a large retrospective study of 606 patients with SHPT undergoing PTX, 1.3% of patients had a cardiovascular event postoperatively: acute cardiac insufficiency in 7 (1.1%), myocardial infarction (MI) in 4 (0.6%), deep venous thrombosis in 1 (0.1%), and a cerebrovascular accident in 1 (0.1%).[266] When compared with the year before surgery, there is a higher incidence of hospitalizations in the year postoperatively for cerebrovascular accident/transient ischemic attacks (relative risk: 1.83%, 95% CI: 1.4–2.4), MI (relative risk: 1.98, 95% CI: 1.6–2.4), and dysrhythmias (relative risk: 1.44, 95% CI: 1.2–1.8).[411] One large retrospective study of 824 patients determined that patients undergoing subtotal resection had a significantly lower rate of cardiovascular events than those undergoing total resection (14.0% vs 29.0%, $P < 0.001$).[258] Single institution studies report a cardiac-related death in 1.8% to 7.5% of patients postoperatively following secondary PTX.[266,426,429] Only one study has reported a cardiac-related death following operation for tertiary disease (1/35 patients, or 2.8%).[342] When studied over longer durations (4–6 years postoperatively), patients receiving PTX for SHPT have a reduced risk for acute coronary syndrome and cerebrovascular event than patients who do not undergo PTX.[430,431]

Mortality

The rate of in-hospital mortality following PTX is up to 3.2%.[265,290,424] These rates are decreasing over time: while the in-hospital mortality rate was 1.2% in 2002, large datasets from contemporary periods now report 0.8% to 0.9% mortality during the index admission.[327,411,422] When expanding to longer postoperative periods, the mortality rate increases with time: 1.1% to 1.5% at 30 days,[320,411] 2% at 90 days,[258] 10% at 1 year,[429] and 12% at 3 years.[290] One study with a median follow-up of 6.6 years postoperatively reported a 40% risk of death,[258] while a second reported that 47% of patients died a mean of 70.7 months postoperatively.[320] When examining by surgical subtype, 2 studies have demonstrated no difference in the mortality rates following subtotal PTX versus total PTX with autotransplant,[257,258] or between total PTX with and without autotransplant.[286] Some of the deaths reported in single-institution studies are unrelated to the surgery, such as deaths secondary to lung cancer[429] or calciphylaxis.[356] Other studies are unable to report on the cause of death due to the nature of the dataset. However, pre-existing cardiovascular disease is a risk factor for mortality, while receipt of a kidney transplant is protective.[258,422] The selection of a high-risk subgroup and metabolic changes following PTX may also play a role.[87]

REOPERATION

Persistent or recurrent HPT are not uncommon in a significant portion of patients who have undergone initial PTX. Persistent disease reflects insufficient resection of parathyroid tissue at the index operation and is usually due to the presence of ectopic and supernumerary parathyroid glands. This outcome occurs even when surgery is performed by an experienced parathyroid surgeon. Recurrent HPT develops in remnant parathyroid tissue due to the ongoing pathophysiologic calcium and phosphorus metabolism in CKD and ESKD. The rate of recurrent HPT after the index parathyroid resection increases with time.[284,432]

There are no standardized definitions of recurrent and persistent HPT. However, there are generally accepted principles

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TABLE 6. Postoperative Calcium and Calcitriol Administration Protocols

References	Ca Monitoring	PO Administration	IV Ca Administration	Discharge Criteria
KDOQI[69]	Ionized Ca q4-6h for 48–72 h, then bid until stable	1–2 gm calcium carbonate tid 2 mcg qD	Begin immediately postoperatively	NA
Cheng et al[420]	Corrected Ca q12h	1 gm elemental calcium qD 0.5 mcg calcitriol qD	If Ca <7.0 mg/dL or symptomatic, add 10 10 mL amps 10% Ca gluconate. Taper when serum Ca stabilizes	When corrected Ca >7.0 mg/dL and asymptomatic
Jain and Reilly[198]	NA	4–12 g elemental calcium daily 2-4 mcg calcitriol daily	NA	NA
Pitt et al[260]	NA	2–3 gm calcium carbonate with or without calcitriol	NA	NA
Saunders[277]	Corrected Ca q4h ×2d	Calcidol/calcichew tablets based upon Ca levels	If Ca <2 mmol/L, begin 10% calcium gluconate IV based upon Ca level	NA
Saunders[166]	Total or ionized Ca q6–8h ×2d	1500 mg calcium TID 1 mcg calcitriol qD	NA	When Ca stable on PO only
Schneider et al[266]		1000 mg calcium ×3 0.5 mcg calcitriol ×3	If Ca <2 mmol/L, give 2gm Ca IV until Ca >2 mmol/L	When Ca >2 mmol/L
Zmijewski et al[421]	NA	800 mg elemental Ca q6h 0.5 mcg calcitriol bid	If Ca 7 mg/dL, begin IV calcium gluconate in doses of 1–5 g	NA

NA indicates not applicable.

which define recurrent or persistent disease. Persistent disease is defined as the failure of PTH to decrease to a certain threshold after surgery, while recurrent disease is defined as initial decrease of serum PTH, followed by a subsequent increase in PTH, generally after 6 months postoperatively. Specifically, persistent disease following total PTX and autotransplantation is frequently defined as a PTH level greater than the upper limit of normal immediately following surgery.[276,284,312,427,433,434] Others have defined persistent disease as a PTH >300 pg/mL immediately postoperatively,[428] a PTH >200 ng/L on 3 occasions within 6 months postoperatively,[298] or a PTH >240 pg/mL within 6 months postoperatively.[320]

Recurrent disease is generally defined as an initially normal (or goal) postoperative PTH level, followed by a subsequent increase in PTH. Six months is considered the threshold beyond which elevation of a previous normal serum PTH is consistent with recurrent disease. Recurrence is defined as a PTH level greater than 3 times the upper limit of normal with an increase in alkaline phosphatase[270]; 3 or more PTH levels >200 ng/L[298]; PTH >300 pg/mL[428]; or PTH persistently >600 pg/mL.[320] Some studies used these numeric thresholds based upon published guidelines for target PTH levels at the time of patient care.[298,312,320]

Regardless of the specific serum PTH levels postoperatively, many agree that hypercalcemia is an important component of the diagnosis of recurrent or persistent HPT in the postoperative setting of THPT.[298] This is because there may still be a component of kidney dysfunction in patients with tertiary disease, which will affect PTH levels and result in inaccurate diagnosis. As such an elevated serum PTH alone is not sufficient for the diagnosis of persistent or recurrent THPT.[298]

Clinical Highlight 9-1: Persistent SHPT is the failure of serum PTH level to fall to a certain threshold postoperatively, and recurrent SHPT an initial decrease in serum PTH level followed by an increase at least 6 months postoperatively. Persistent and recurrent THPT are diagnosed with hypercalcemia in the setting of elevated or inappropriately normal PTH level.

Causes of Persistent and Recurrent Disease

Persistent HPT usually indicates the presence of one or more unresected glands. After total PTX with autotransplantation, persistent disease is almost never due to the parathyroid autograft; similarly, persistent disease after subtotal PTX is unlikely to be the result of too large a remnant.[270,427,433] In this scenario, fewer than 4 glands may have been identified during the index operation, therefore 1 or more glands may not have been resected. Supernumerary glands may also cause persistent disease. When present, ectopic and supernumerary glands are commonly located in the thymus, mediastinum or carotid sheaths, in the thyroid, along normal cervical fatty tissue in the tracheoesophageal groove, posterior to the esophagus, or are undescended and higher in the neck.[287,312,435]

Recurrent disease following subtotal PTX can be due to (1) regrowth of the remnant gland; (2) stimulation or growth of a parathyroid rest; or (3) parathyromatosis.[270,433] Recurrent disease following total PTX can be due to (1) stimulation of growth of a parathyroid rest; (2) parathyromatosis, or (3) stimulation of the autotransplant.[433,435] Parathyromatosis is caused by the seeding of parathyroid cells into the operative field, possibly due to improper handling of the gland.[270] The parathyroid tissue

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becomes densely adherent to the surrounding structures, making re-excision difficult.[434] Because of the nature of SHPT, in which the capsule of one parathyroid gland is intentionally ruptured to create the remnant, parathyromatosis is more common after subtotal resection.[435] Multiple operations may be needed to treat parathyromatosis, and the prognosis is poor, although calcimimetics can be used.[270,434,435] This process can also occur at the site of an autograft.[434]

Clinical Highlight 9-2: Persistent HPT is due to the presence of unresected parathyroid glands, while recurrent HPT is due to growth of remnant glands or parathyroid cells.

Criteria and Decision-making to Reoperate

Reoperation is associated with higher complication rates and operative failure rates compared to the initial PTX. Consequently, patients with persistent or recurrent HPT may experience delayed referrals to surgery.[435] A variety of criteria have been proposed to objectively define which patients are likely to benefit from additional surgery. Unfortunately, many authors do not distinguish between the definitions of persistent and recurrent HPT, or the indications to reoperate. Hence the decision can be challenging. However, most providers agree that a combination of PTH, calcium and phosphate levels can guide the decision to reoperate on patients with persistent or recurrent HPT. Examples of such cutoffs are described below.

PTH levels can be used to indicate a need for reoperation. PTH levels >500 pg/mL[427] and >800 pg/mL[428] have been proposed as cut-offs for reoperation. Others use the presence of three PTH levels above the upper limit of normal in conjunction with three elevated calcium levels.[293] Other biochemical criteria for reoperation include hypercalcemia.[428,432] In addition, some surgeons factor the presence of signs and symptoms into the decision to pursue reoperation.[293] In particular, osteitis fibrosis cystica or other high bone turnover refractory to medical management is considered an indication to reoperation.[427,433,435] One group has biochemical or imaging requirements in addition to the presence of bone pain, skeletal deformities, anemia, or pruritis.[428]

Radiologic imaging findings can also be used as criteria for reoperation. The presence of parathyroid glands seen on 2 or more imaging modalities[296,427] and gland hyperplasia with a diameter of >0.5 cm and increased blood flow on ultrasound have both been reported.[428] No studies have been performed to compare the utility of the various criteria to reoperate. However, all patients being considered for reoperation should have disease that is refractory to medical therapy.[284,428,432,435]

Recommendation 9-1: Reoperation should be considered for patients with medically refractory persistent or recurrent HPT. (Strong recommendation, moderate-quality evidence.)

Imaging and Preoperative Localization

In contrast to an initial PTX, preoperative localization is essential prior to reoperation in SHPT or THPT due to the many possible locations for hyperfunctioning parathyroid tissue. In the setting of persistent disease, surgeons will benefit from preoperative identification of a missed ectopic or supernumerary gland in the neck or mediastinum. In the setting of recurrent disease, surgeons need to identify a possible hyperplastic remnant gland, the presence of supernumerary glands, and any parathyromatosis; following autotransplantation, surgeons will also need to evaluate the autograft for hyperfunction. Imaging studies, while not always definitive in initial operations for SHPT or THPT, can be helpful to localize enlarged parathyroid glands in the reoperative setting. Sestamibi scintigraphy has reported sensitivity of 71.0% to 95.2%[270,296,436,437] and positive predictive value of 100% in patients undergoing reoperation.[435] In contrast, ultrasound has a lower sensitivity of ~50% to 71.4%.[270,436] This may be due to the high rate of ectopic glands, in particular mediastinal glands, that may not be readily identified on ultrasound. Cross-sectional imaging can be used to identify ectopic glands and may assist in providing anatomic landmarks for the surgeon. A meta-analysis reported that computed tomography has a true-positive rate of identifying parathyroid glands of <50%, and magnetic resonance imaging has a true-positive rate of 54%.[436] More recent data reflecting technological improvements is unavailable. Consequently, some authors advocate for the combination of ultrasound and scintigraphy as the gold standard noninvasive preoperative imaging study in patients requiring reoperation.[436] In parathyromatosis, preoperative localization is challenging. Scintigraphy can demonstrate small foci of uptake throughout the neck.[435] Ultrasound is likely to demonstrate small nodules around the thyroid lobes.[434]

In patients with an autotransplant, the autograft site is the most common site of recurrence.[295,298] Notably, all published literature on imaging a graft site discusses solely the forearm. The forearm can be imaged with ultrasound, sestamibi, computed tomography or magnetic resonance imaging.[293,296,434] When the forearm is the autograft site, blood draws can also be used to identify the site of recurrence. While comparing the ratio of serum PTH between the grafted and nongrafted arms is commonly performed, looking for a gradient of >1.5[432] or >2,[296] many surgeons perform the Casanova test or a modification of it. The Casanova test consists of placing 2 tourniquets proximally on the grafted arm and inflating the proximal cuff to >100 mm Hg. Ischemia is maintained for 30 minutes, and blood samples from both arms are then at 10, 20, and 30 minutes of ischemic time. Ten minutes after return of flow, blood samples are again drawn. A PTH decrease of $>46\%$ in the nongrafted arm during ischemia and a return to basal levels after conclusion of the ischemic time indicates that the graft is the site of the hyperfunctioning tissue. In contrast, a decrease of <20% indicates a different source.[294] A modification of the test draws only from the nongrafted arm, looking for a ratio of $>60\%$ between the ischemic draws and the basal.[427] One study has compared the Casanova test to ultrasound and sestamibi. This study concluded that the Casanova test (87.5%) was more sensitive than scintigraphy (68.7%) and ultrasound (56.2%).[293]

There are no common or established algorithms for choosing localization modalities. The choice of study depends both on the type of initial operation and if the patient has persistent or recurrent disease. If the noninvasive studies described above fail to localize disease, selective venous sampling (SVS) can be performed.[438] SVS has a higher sensitivity[354] and accuracy than both ultrasound and sestamibi, but lower specificity.[439] However, SVS is invasive, requires both increased labor and specific technical expertise, and has the potential for adverse reactions such as kidney failure and contrast dye reactions.[439] Consequently, SVS is not the first choice for diagnostic imaging but can be useful if noninvasive imaging studies are indeterminate or inconclusive.[439]

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Recommendation 9-2: Preoperative localization is crucial prior to undertaking reoperation for recurrent or persistent kidney-related HPT. The choice of localizing imaging or procedure depends on the type of initial operation and local expertise. (Strong recommendation, low-quality evidence.)

Reoperation Following Subtotal PTX

Imaging should be performed preoperatively, as discussed above, in order to localize parathyroid glands in the neck. Once in the operating room, general anesthesia is induced, and the patient is positioned and draped as for an initial PTX. Intraoperative nerve monitoring should be employed to assist with identification and protection of the RLNs.[428] The neck can be approached through the same incision[428] or a lateral approach; if a gland has been identified in the mediastinum, sternotomy may be indicated.[427] Once the central neck has been entered, a bilateral exploration should be performed in order to identify all remaining parathyroid tissue, including the remnant gland as well as any ectopic or supernumerary glands.[428] The goal of the operation is to remove as much parathyroid tissue as possible without rendering the patient hypoparathyroid. There is no evidence supporting either total PTX with autotransplantation versus subtotal resection in this setting. Surgeons performing subtotal PTX with preservation of a remnant should be conscious of the possibility of devascularization of the remnant during the exploration, thus potentially rendering the patient hypoparathyroid.[433] If there is concern for devascularization of the remnant, immediate autotransplantation should be performed.[440] Total PTX with autotransplantation is the operation of choice for other surgeons,[271,295,441] ideally with cryopreservation.[271] Thymectomy is indicated to resect any parathyroid tissue within the thymus.[428] If a parathyroid gland is intrathyroidal based upon preoperative imaging, a thyroidectomy should be performed.[428] If parathyromatosis is identified, an en bloc resection consisting of a thyroid lobectomy, ipsilateral lymph node dissection and excision of surrounding adherent fibrous tissue may be performed to facilitate resection; however, complete resection can be difficult to perform.[434] Frozen section can be used to confirm the presence of parathyroid tissue in the resected specimens.[428] There is no published literature on the performance of ioPTH monitoring in the reoperative setting.

Recommendation 9-3: Reoperation following subtotal PTX should identify all remaining parathyroid tissue and either subtotal PTX or total PTX with autotransplantation should be performed. (Weak recommendation, low-quality evidence.)

Reoperation Following Total PTX and Autotransplantation

If overproduction of PTH stems from both the autograft and the neck or mediastinum, a graftectomy is initially performed to avoid reoperation in a scarred field.[427] Removal of the autograft can be performed under either local or general anesthesia.[293,298,428,432] The same incision is used. The nonabsorbable suture placed at the index operation identifies the implantation site(s).[293] All identifiable parathyroid tissue should be removed to prevent recurrence.[270,293,432] Parathyroid tissue can be difficult to locate because the tissue may be embedded into the surrounding muscle. Consequently, an en bloc resection of the surrounding fascia, muscle and soft tissue is required.[295,428,432,433] Frozen section can be utilized to confirm the presence of parathyroid tissue,[293] which should be cryopreserved.[270,296] Because of the difficulty in completely resecting autografted parathyroid tissue, multiple debulkings are required in up to 50% of patients.[295]

Reoperation in the neck is indicated due either to persistent disease, or recurrent disease localized solely to the neck or not responsive to initial grafetectomy.[427] When reoperation in the neck is needed, surgery should proceed as described above for reoperation following subtotal PTX. If the autograft is intact, no new autotransplant is indicated.[427] Only one study has published an ioPTH monitoring protocol for resection. This study, which described solely autograft removal following initial total PTX and autotransplantation, reported a goal of >50% decrease at 10 minutes following resection and >60% at 15 minutes.[293]

Recommendation 9-4: Reoperation following total PTX with autotransplantation depends on the site of persistent or recurrent disease. If indicated, graftectomy should be initially undertaken. (Weak recommendation, low-quality evidence.)

Definition of Cure

As with initial surgery for SHPT, there is no consensus on the definition of cure following reoperation. A PTH target is often employed, although the goal level or range varies. Surgeons aim for a normal or low PTH level; a level less than twice the upper limit of normal; 60 to 240; 150 to 300; or <300 pg/mL.[312,428] Some of these target ranges reflect published guidelines for goal PTH levels at the time of patient care.[312,438] One group also factors in improvement of associated symptoms (bone pain, pruritus, muscle weakness) in additional to a low or normal PTH level.[433]

Clinical Highlight 9-3: There is no commonly accepted definition of cure following reoperation for SHPT.

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